Microsoft Power Platform

Lifecycle Mastery: Managing Power Platform Assets in the Age of Agents

Get the labs

https://aka.ms/PPCC25-ALM

Microsoft Power Platform

Presented by:



Marcel Ferreira
Principal Product Manager
Developer experiences



Prabhat Pandey
Principal Engineering Manager
DevOps



Casey Burke
Principal Product Manager
DevOps



Yash Patel
Senior Program Manager
Power CAT



Ashish Choudhary
Principal Engineering Manager
ProDev



Priyanshu Agrawal
Senior Software Engineer
ProDev

Microsoft Power Platform

Proctors in the room:



Brian Gregor
Power Platform
Global Black Belt



Olivier Conort
Power CAT
Microsoft



Henry Luo
Power CAT
Microsoft



Ravi Chada
Power CAT
Microsoft

Get the labs

https://aka.ms/PPCC25-ALM

Agenda

Lab	Time	Topic
Presentation	9:00 AM	Solutions and DevOps 101
Lab 1	9:20 AM	Setup and configure
Lab 2	9:40 AM	Building Power Platform Solutions
Coffee	**10:00 AM**	**Coffee break**
Lab 2 (cont)	10:15 AM	Building Power Platform Solutions
Lunch	**11:45 AM**	**Lunch**
Lab 3	12:45 PM	Deploy
Lab 4	2:15 PM	Lifecycle Mastery in Power Pages
Presentation	3:00 PM	Best practices & Q&A

Microsoft Power Platform

Solutions and DevOps 101

Show of hands:

Are you new to DevOps (ALM) in Power Platform?

Show of hands:

< 2 years experience with DevOps in Power Platform?

Show of hands:

2-5 5 years experience with DevOps in Power Platform?

Show of hands:

5-10 years with DevOps in Power Platform?

Show of hands:

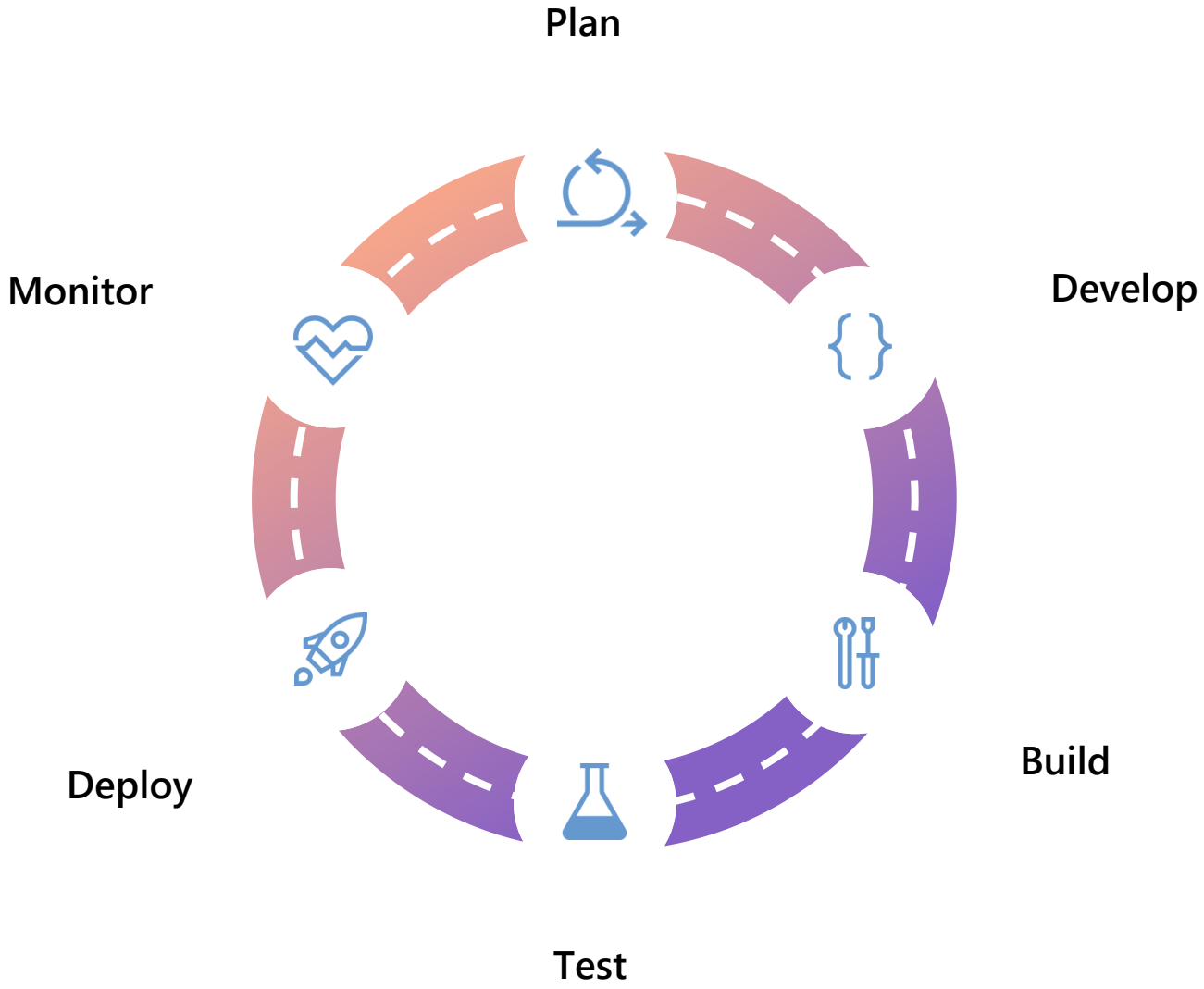
10+ years with DevOps in Power Platform?

3x usage

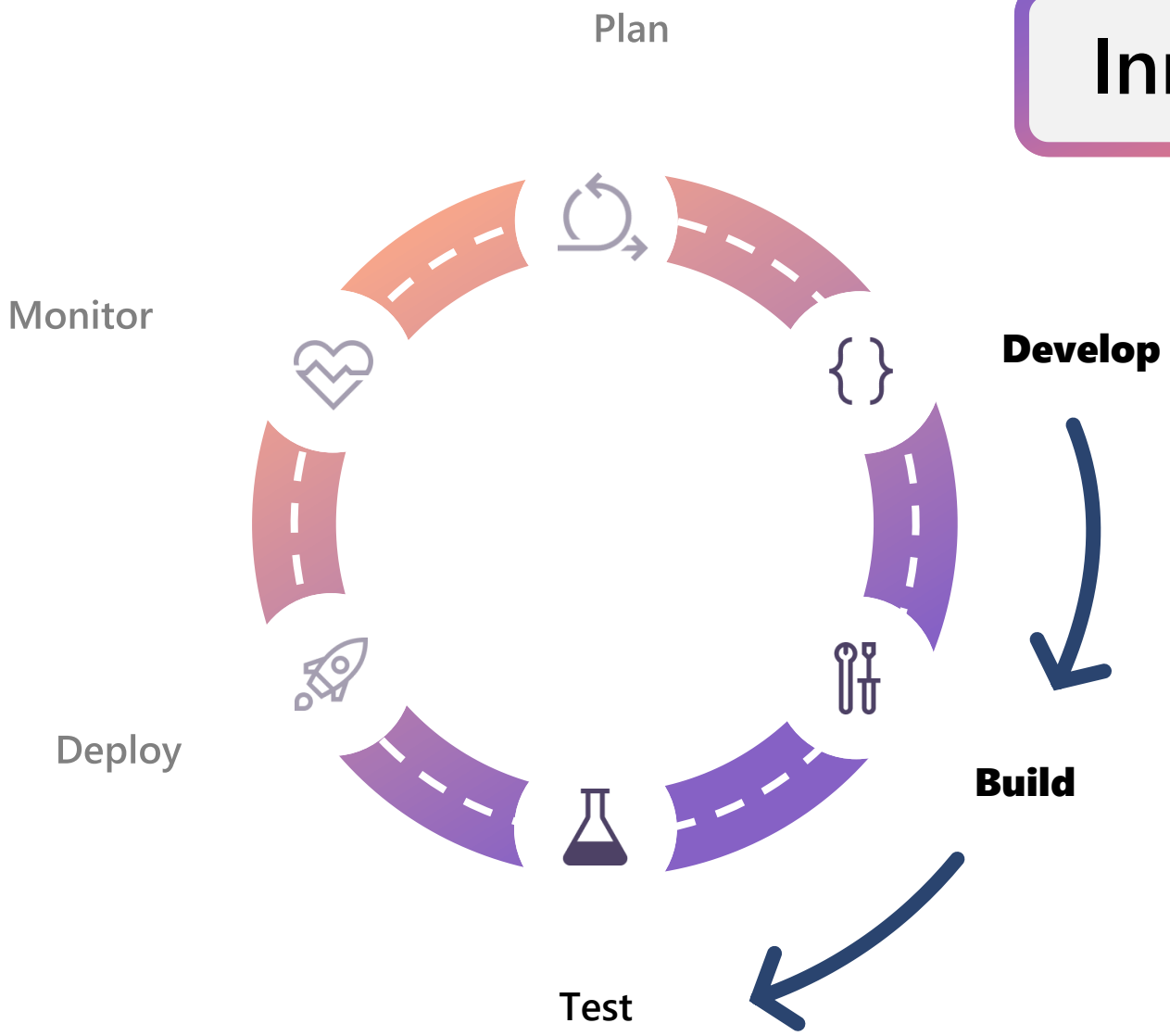
Solutions using DevOps (ALM) have 3x more usage on average

What is DevOps? (aka ALM)

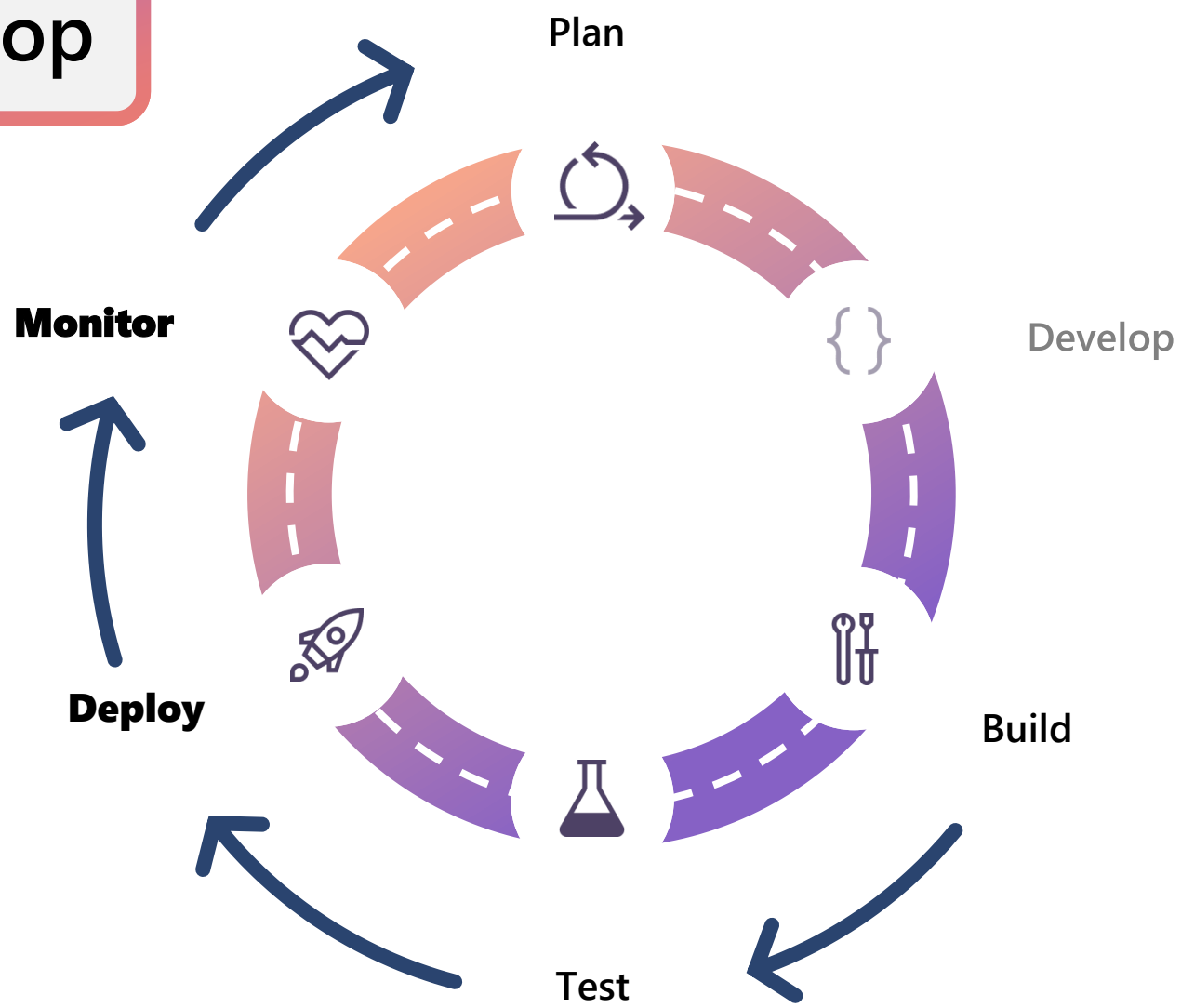
*DevOps is the union of **people**, **process**, and **products** to enable **continuous** delivery of value to end users*


Plan
Develop
Build
Test
Deploy
Monitor

Inner Loop



Outer Loop



DevOps fundamentals in Power Platform

Environments

Definition:

Container to store, manage, and share your organization's business data, agents, apps, automations and other Power Platform resources.

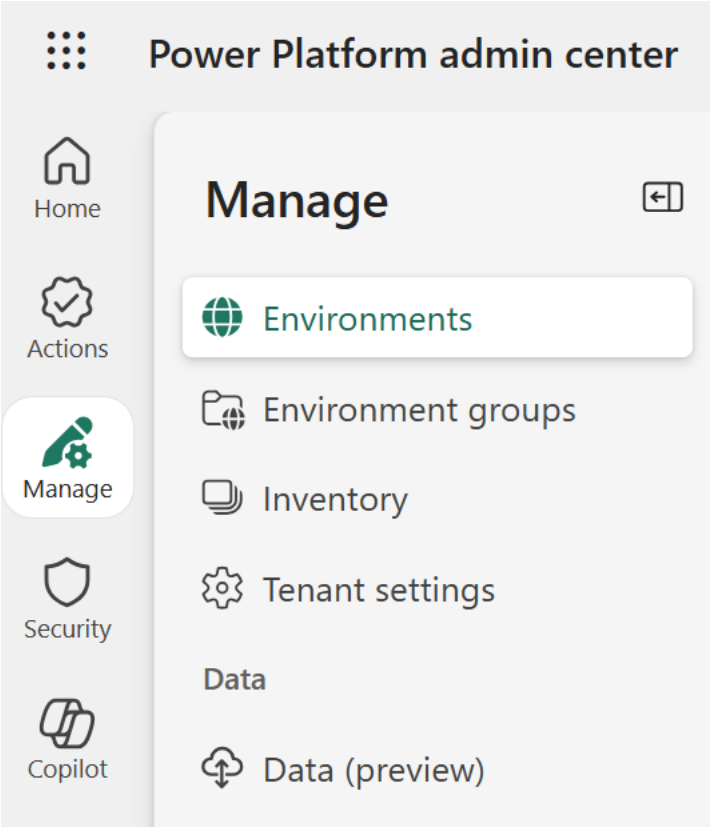
Serves as an isolation barrier to separate applications and data that might have different roles, security requirements, or target audiences.

Key facts:

- Tied to a geographic location
- Different types of environments exist
- Can have either **zero** or **one** Dataverse database
- Every tenant has a Default environment where all makers can create.
- Environments can be grouped together to apply policies uniformely

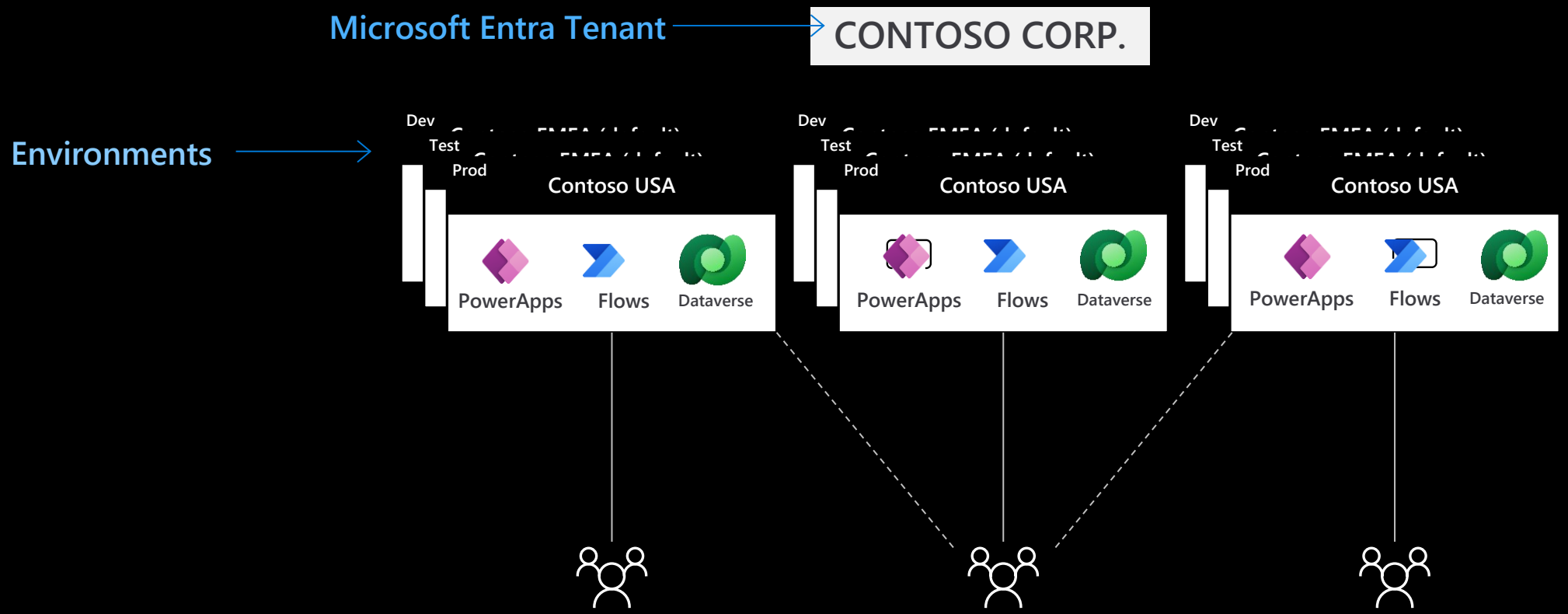
Security tips:

- Non-default environments offer more permission controls.
- Non-default environment creation can be restricted to specific users.
- Test and production environments should use least privileged access

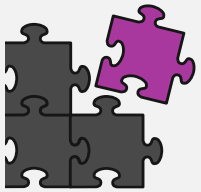


Develop Your Environment Strategy

Environments are containers that administrators can use to manage agents, apps, automations, Pages, and other assets; along with permissions to to use the resources

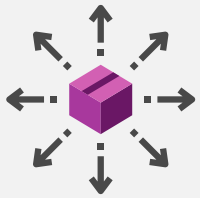


What is a solution?



A ***Solution*** is a **group** of one or more **related objects** that make up an **“application”**

A solution **publisher** defines **ownership** of the objects within a solution



Solutions allow ***distribution*** of applications across environments

What goes into a solution?

Data	User Interface	Analytics	Process / Code	AI	Security	System Settings
• Tables • Columns • Relationships • Choices • Connectors	• App • Forms • Views • Command bar • Controls • Pages	• Dashboards • Reports • Charts	• Workflows • Plugins • Assemblies • Plugin Steps	• Agents • AI models	• Security Roles • Column Level Security Profiles	• Formatting Options • Other System Settings

Solutions contain your customizations of anything in Power Platform, but not business data.

What are unmanaged and managed solutions?



Unmanaged Solution

Only for **development** of the solution

Collection of references to objects
Can be exported as unmanaged or managed



Managed Solution

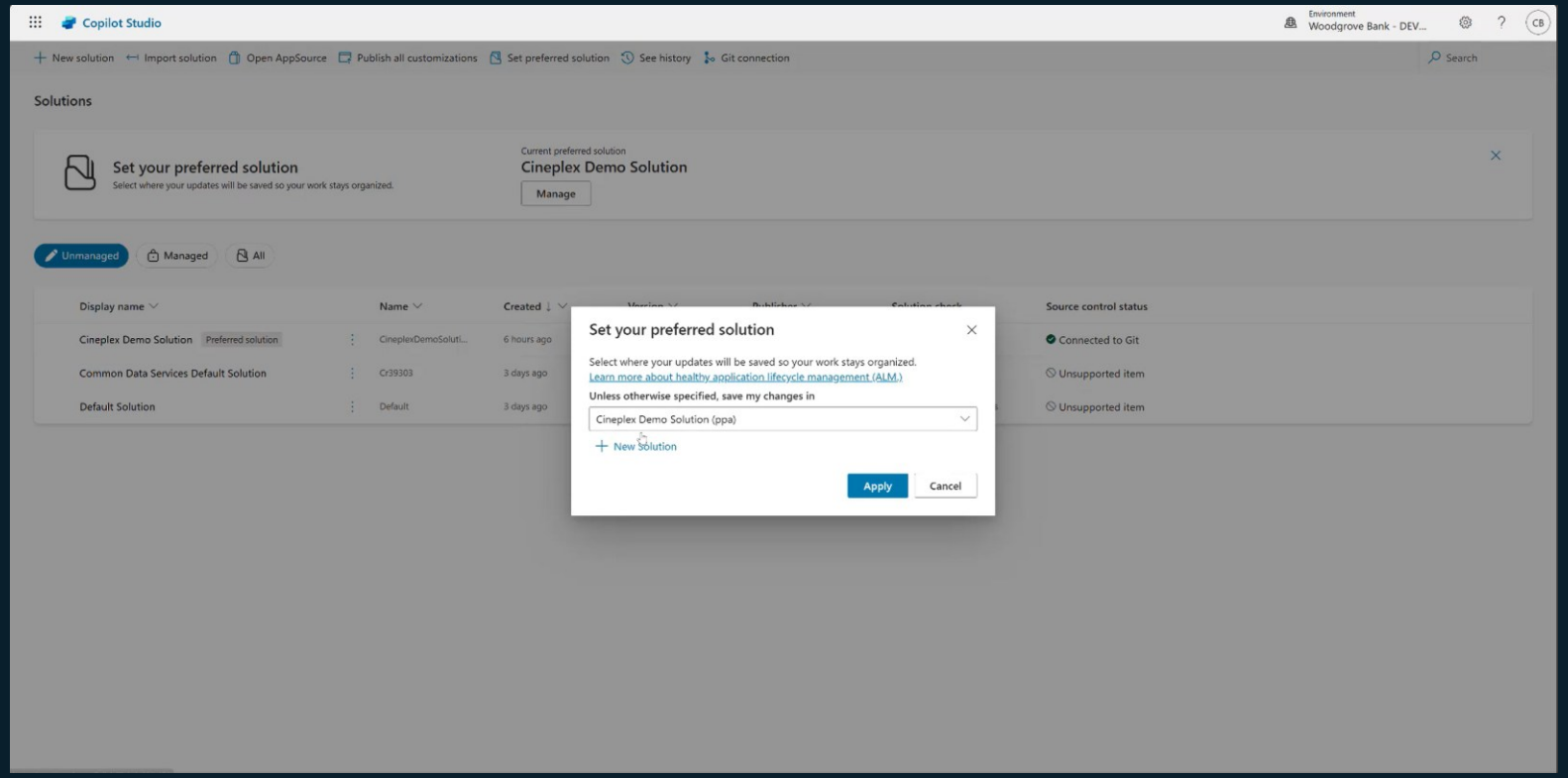
Only for **deployment** of a solution

Sealed “binary”
Components can’t be added or removed
Cannot be exported

Preferred Solution

Highly recommended

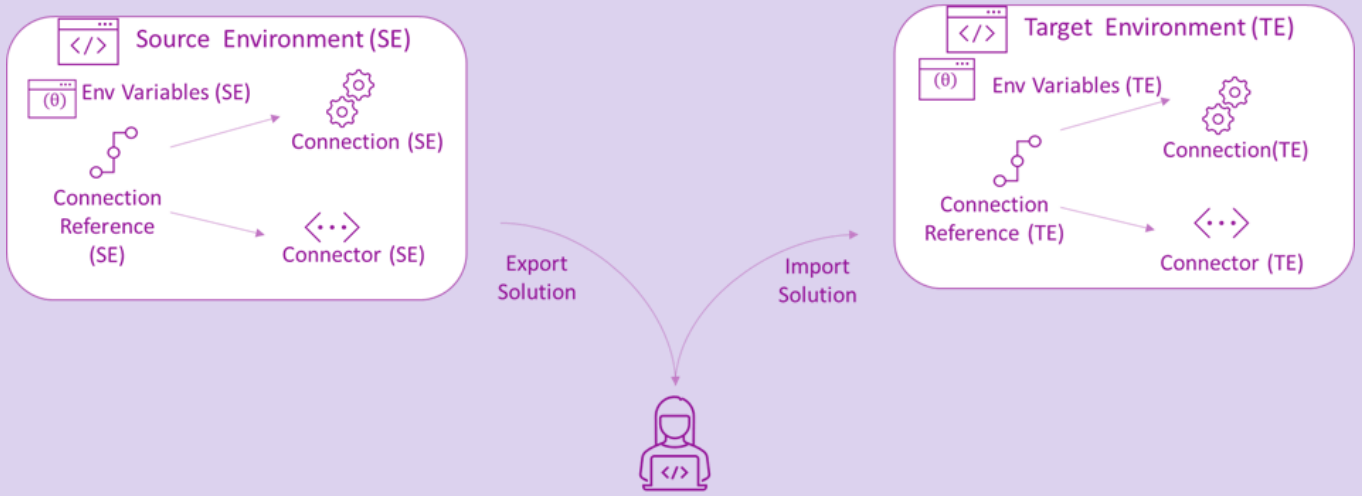
- Starts every maker with “healthy” ALM
- Organizes customizations automatically
- Works across all maker portals



Environment variables and connection references

- Environment variables allow you to transport application configuration data with solutions and optionally manipulate the values in your Application Lifecycle Management (ALM) pipeline.

- Connection references ensure that connections can participate in ALM by removing hard-coded identifiers within components and allowing new connections to be associated as you migrate solutions across environments.



Tools you'll use

Plan
Plan designer
Other 1st and 3rd party tools

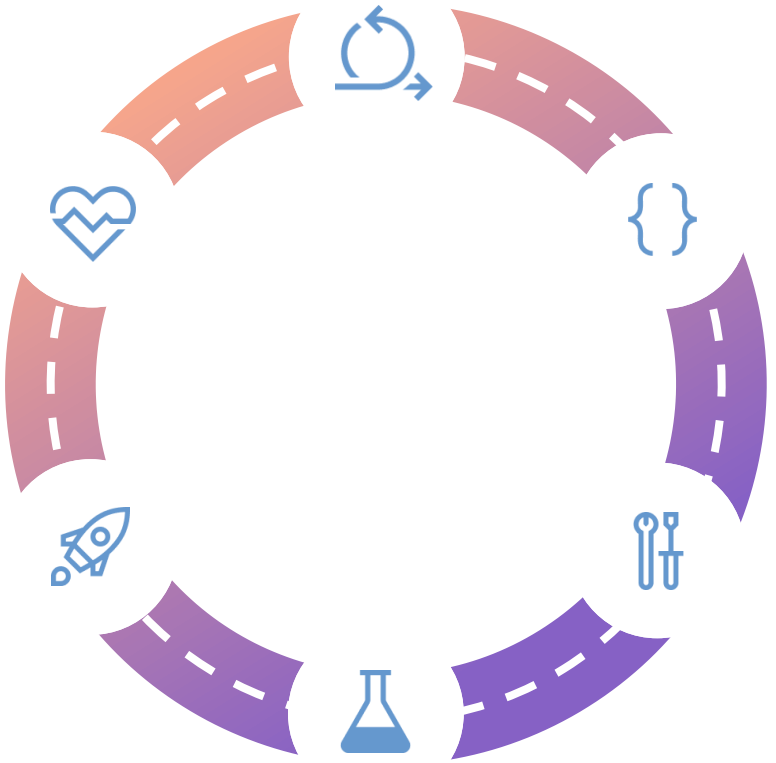
Develop
Maker experiences
Native source code integration
1st and 3rd party developer tools

Build
Pipelines in Power Platform
1st party developer tools

Test
Maker experiences
1st or 3rd party test frameworks

Deploy
Pipelines in Power Platform
Developer tooling

Monitor
Monitor Page
Other 1st & 3rd party tools



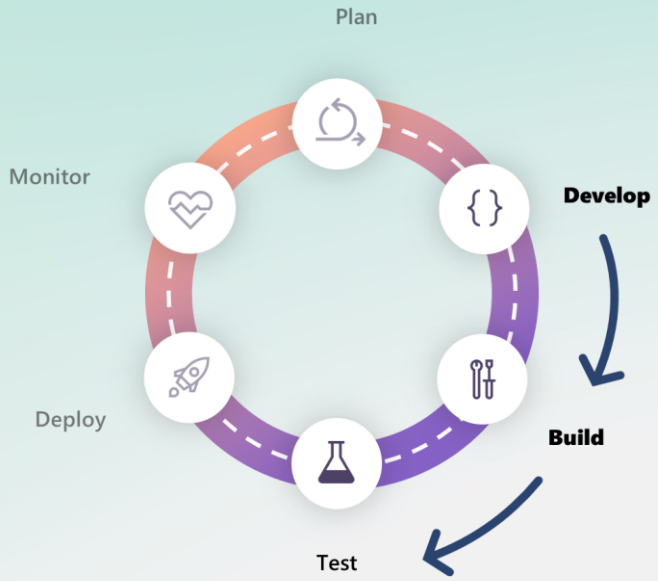
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Lab 1: Setup and Configure

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Lab 2: Building Power Platform Solutions

The inner loop



Why source Code matters?

6. Assembly

Code:

```
MOV AX, 0       ; sum = 0
MOV CX, 1       ; i = 1
LOOP_START:
ADD AX, CX      ; sum += i
INC CX
CMP CX,
JL LOOP_
```

1. Copilot

Prompt:

"Calculate the sum of numbers from 1 to 10"

```
1  int sum = 0;
2  for(int i = 1; i <= 10; i++
3      sum += i;
```

2. Python

Code:

```
sum(range(1, 11))
```

3. C# (Object-Oriented)

Code:

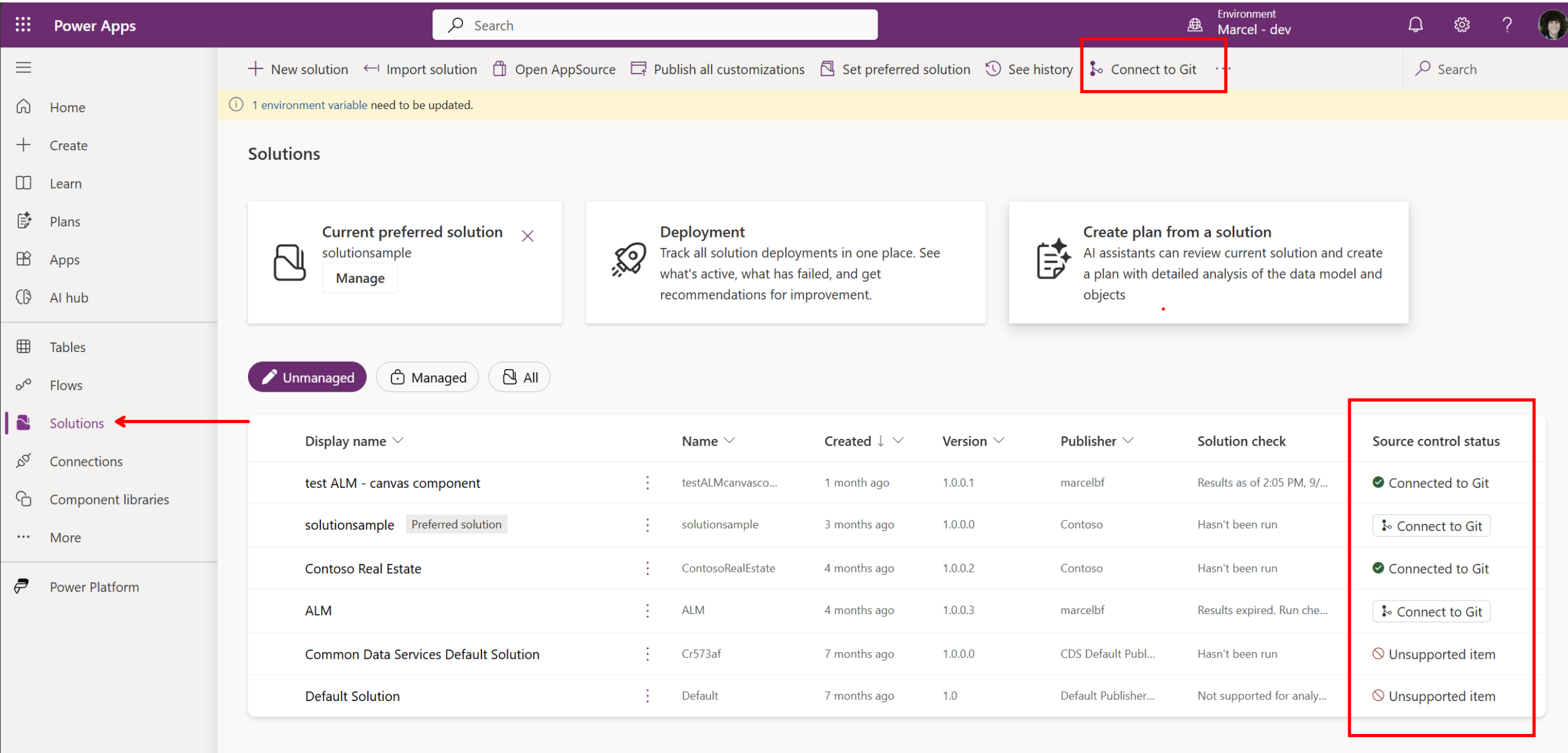
```
1  class Calculator {
2      public int Sum(int start, int end) {
3          int sum = 0;
4          for (int i = start; i <= end; i++)
5              sum += i;
6          return sum;
7      }
8  }
```

4. F# (Funct

Code:

```
let sum = [1..10] |> List.sum
```

Solution is the Source Code



Working with solution files (.zip)

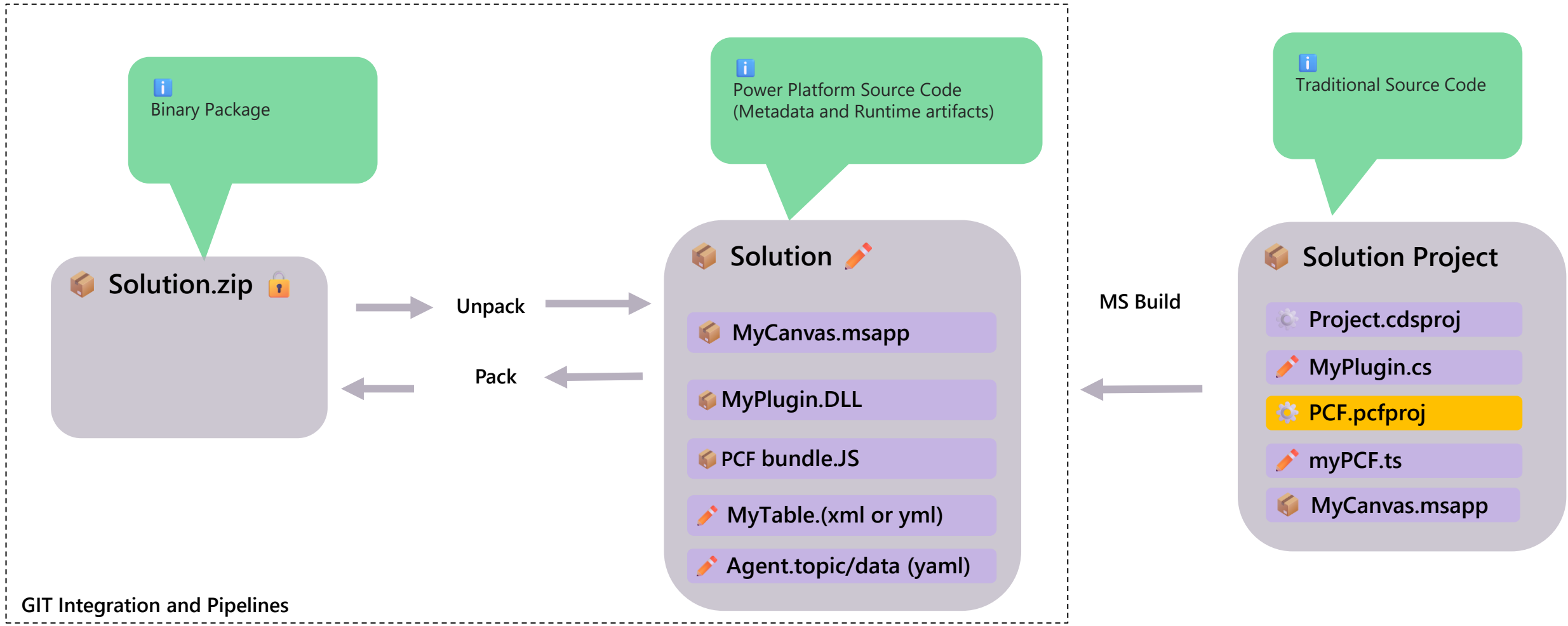
LAB 2 - Task 1

estimated time: 5 min

Expected outcome

- Your solution is imported in your environment
- Your Dev environment is connected to your repository
- Your solution is committed to your source control.
- You can see the unpacked solution in your repository

Solutions Source Code



Working with source code

LAB 2 - Task 2
step 1 and 2.

estimated time: 10 min

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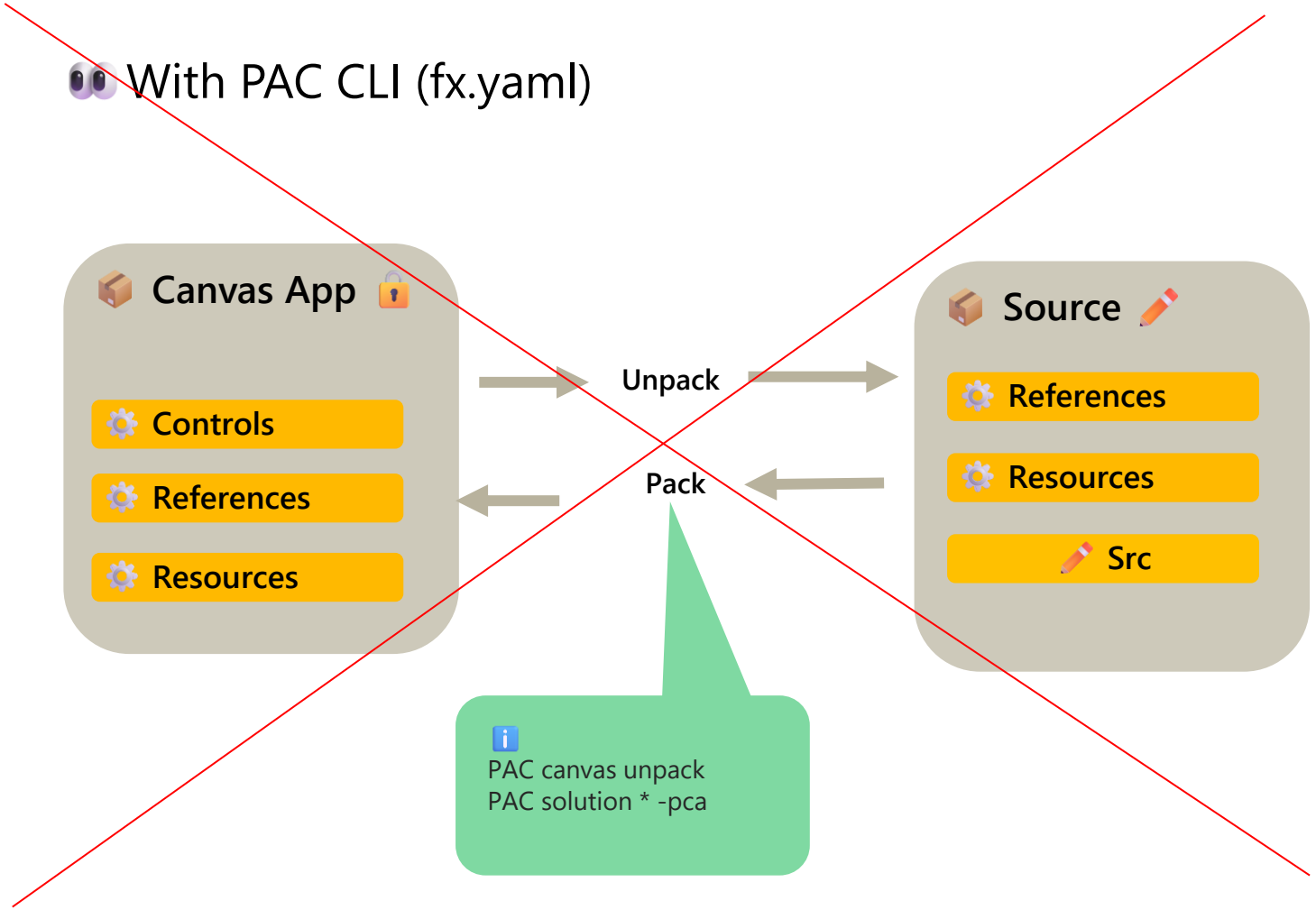
Coffee Break 15 min

Agenda

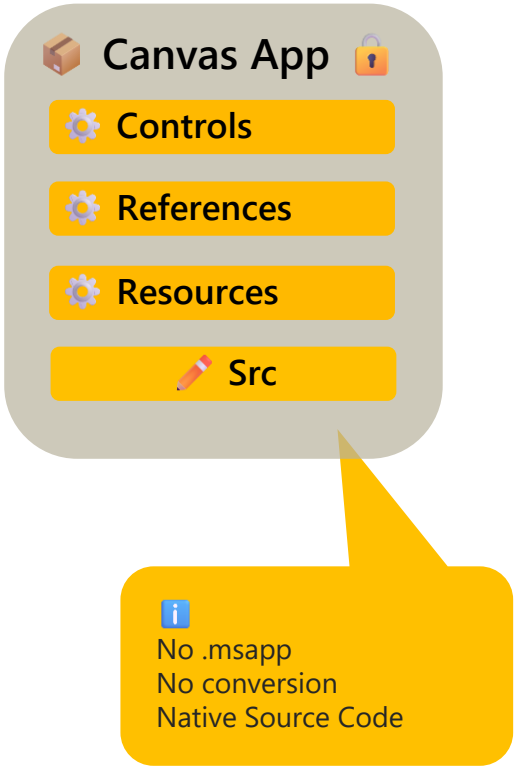
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How to Source Control

With PAC CLI (fx.yaml)



With Dataverse GIT Integration (pa.yaml)



Microsoft Confidential

Working with source code

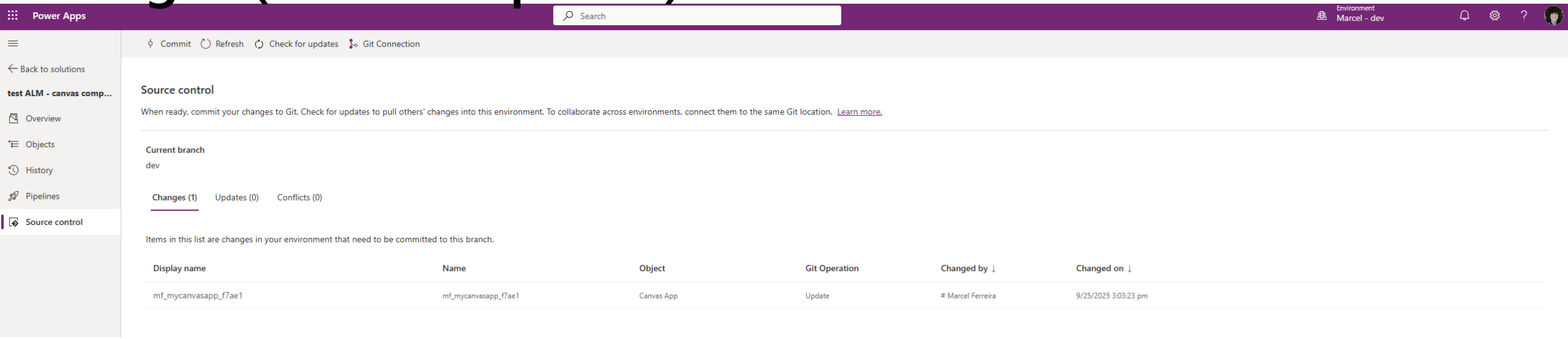
LAB 2 - Task 2

estimated time: 10 min

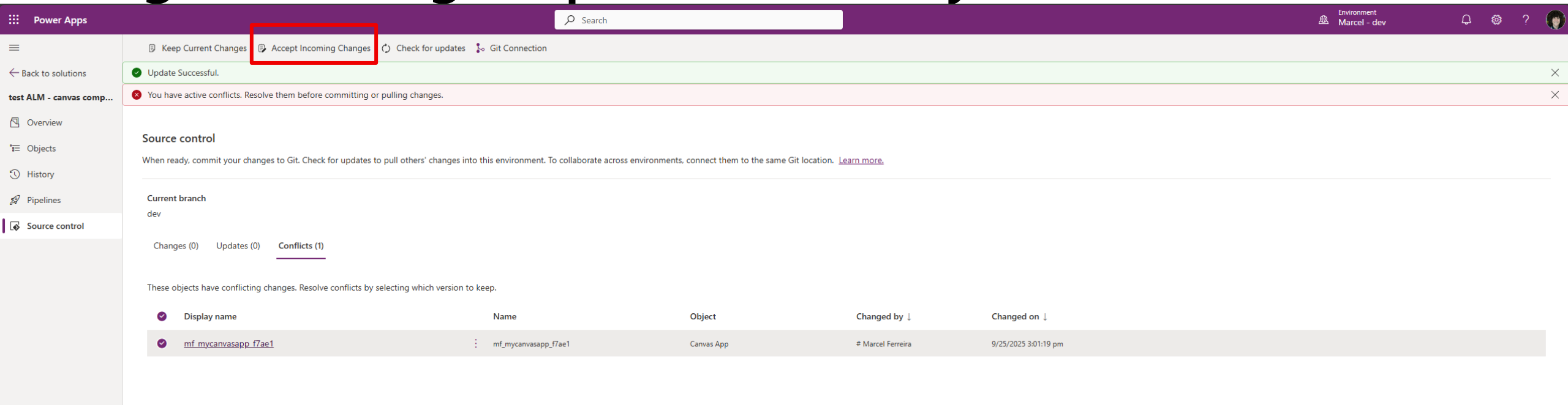
Expected outcome

- You know how to commit in maker portal.
- You can find your source code repository in Azure DevOps
- You know how to get updates from your repository to your environment.
- You understand how to handle conflicts

Changes (in maker portal)



Changes (in the git repo) - Reverse sync



Working as a Team in multiple environments

Branching

Main branch is protected



Main branch represents production and can only be merged via PR after deployment

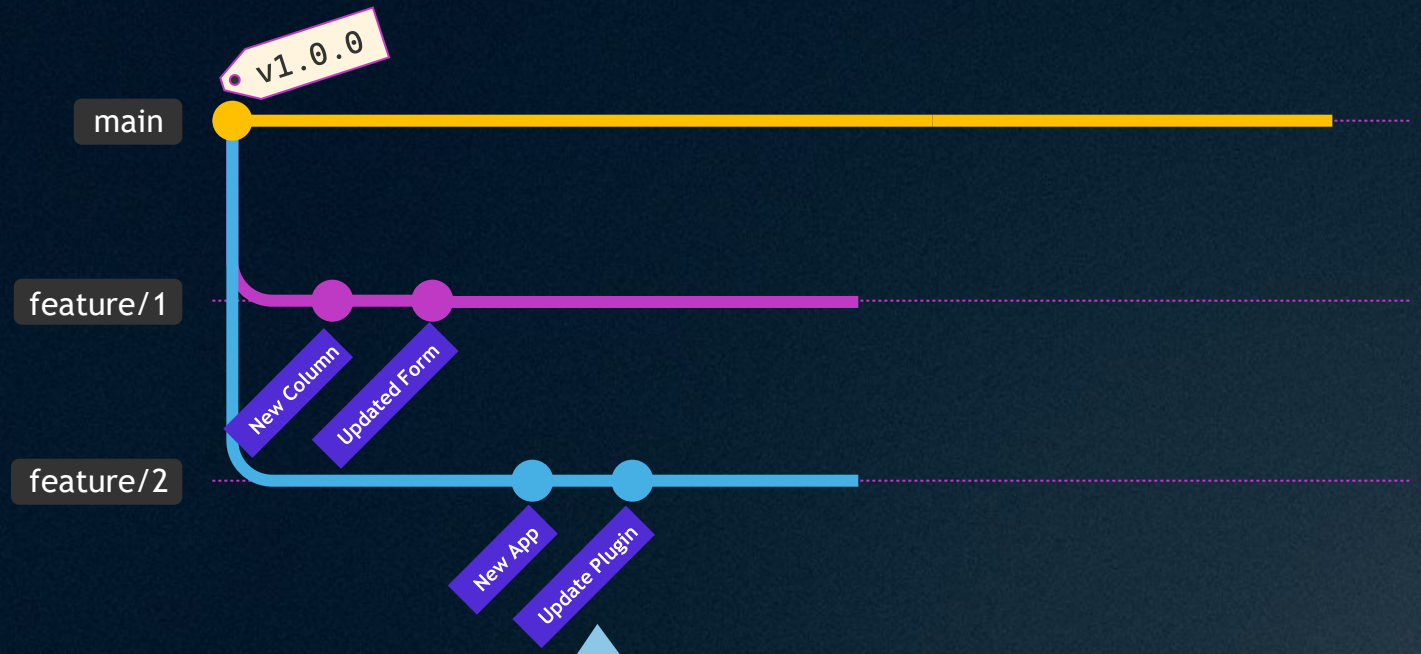
Tags represent the released version

Feature 1 branch and environment



Feature 1 Development starts on a branch of main

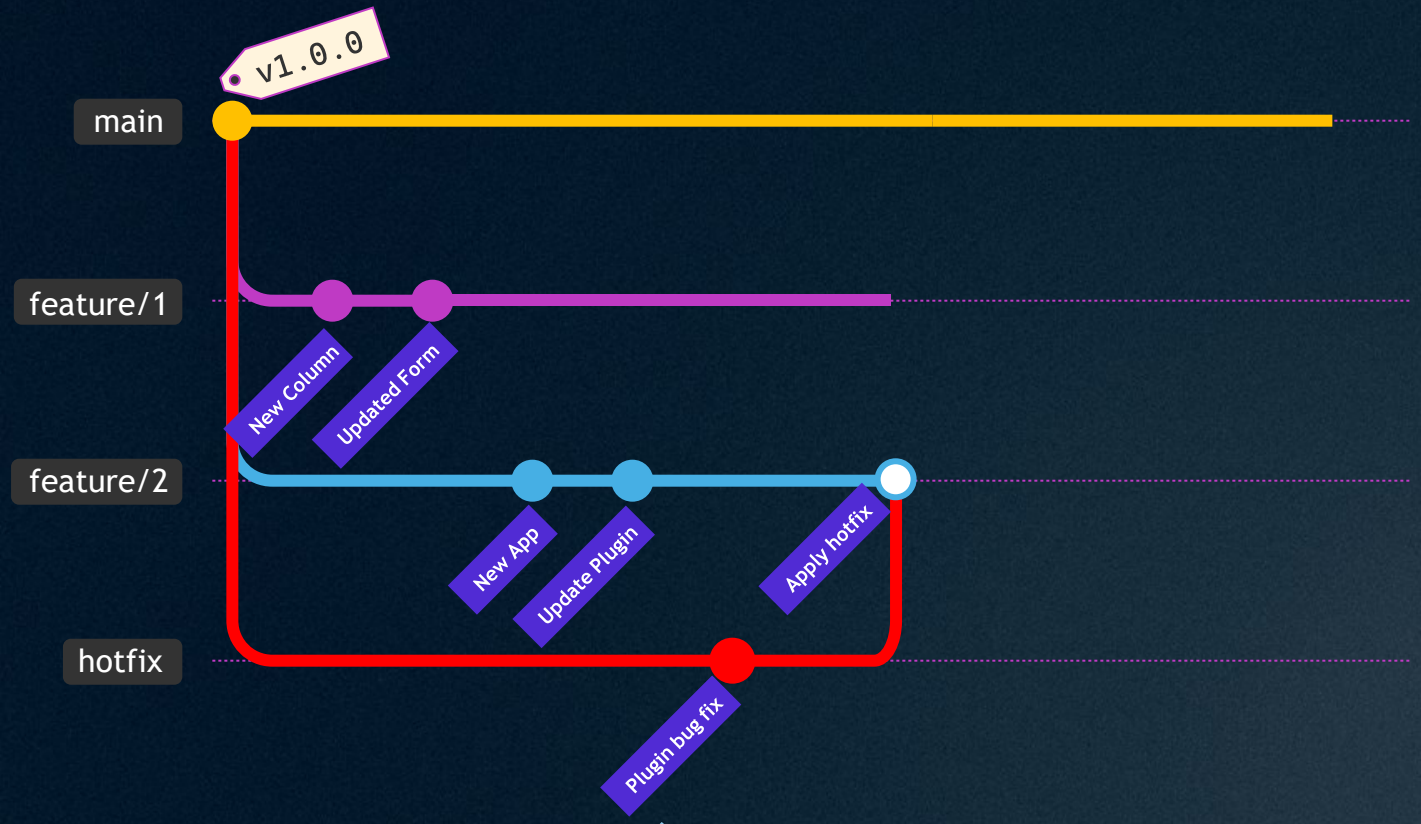
Feature 2 Branch and Environment



Feature 2 will likely be released after feature 1

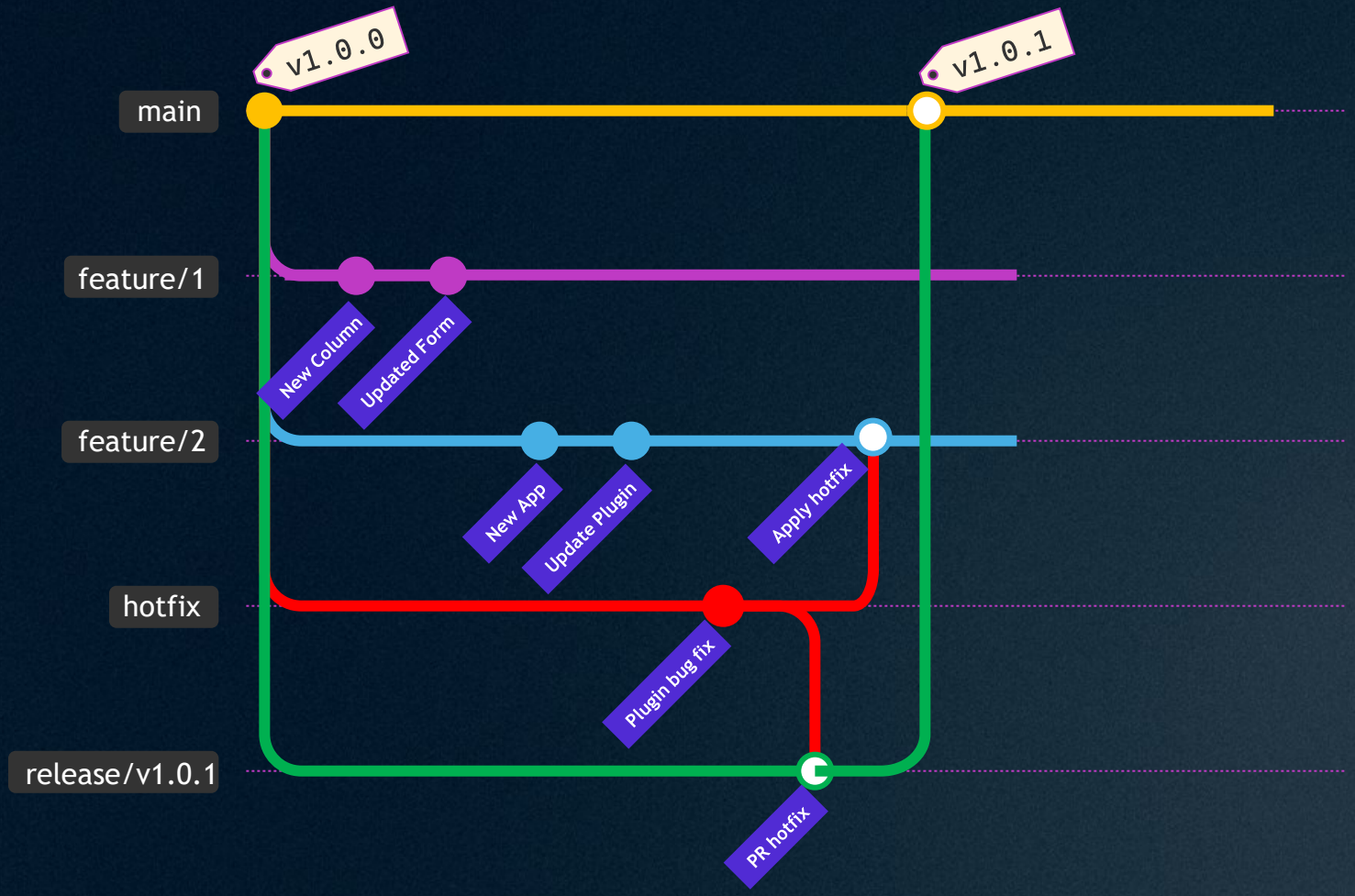
Feature 2 development starts in parallel on a branch of main in a **NEW ENVIRONMENT**

Hotfix created



A bug is discovered in production, and a hotfix is created in a branch of main in a **NEW ENVIRONMENT**

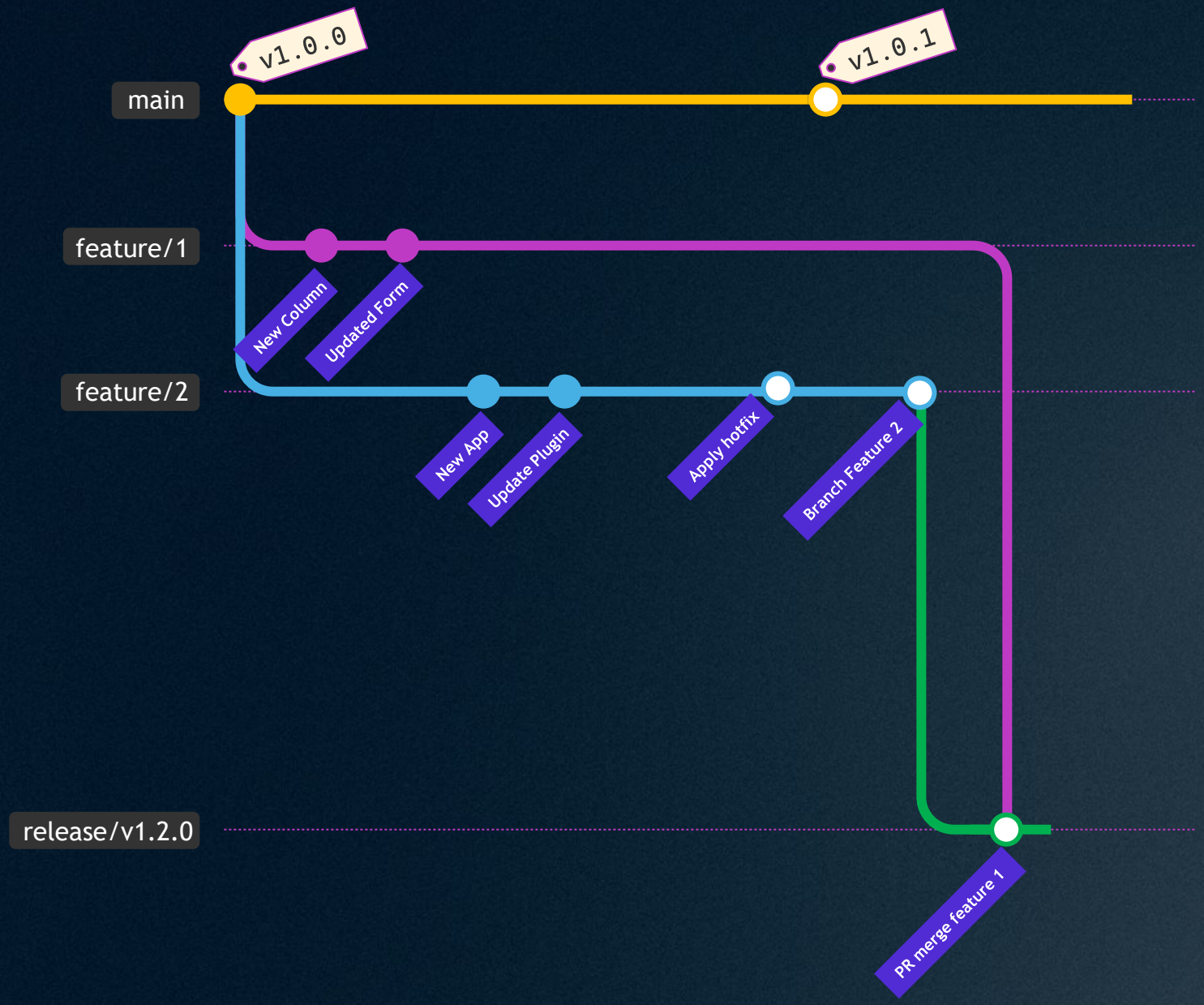
Hotfix Released and merged



A Release branch and NEW ENVIRONMENT is created, and the hotfix is merged in, and deployed. The hot fix is merged back into feature 2 because it effects the same code.

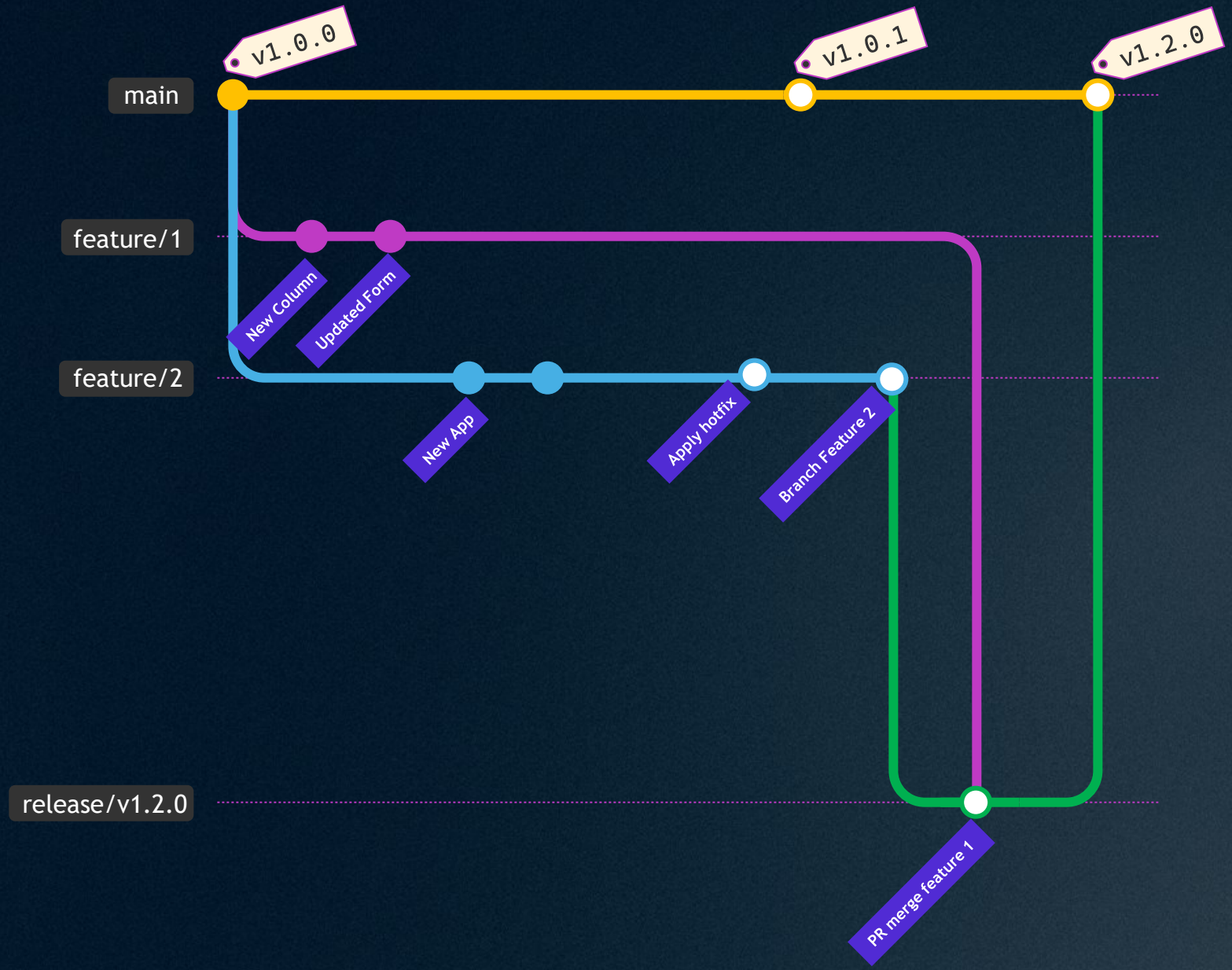
Version 1.0.1 hotfix is released and merged to main

Branch for release



Version 2 branch is created from feature 2 that contains the hotfix, and deployed to a new environment

Deploy V2



Feature 1 is merged in. No merge conflicts should exist. If they do, the **changes must be back-ported** to feature 1. Version 2 is deployed.

Branching and merge

Lab 2 - Task 3

estimated time: 20 min

Expected outcome

- Understand the 1:1 environment to branch relationship
- Created a hotfix environment / branch
- Created the hotfix in an isolated environment
- Merged the hotfix branch in your main branch to apply the hotfix.

Agents

LAB 4

estimated time: 20 min

Expected outcome

- You have created a Copilot Studio Agent within a solution.
- Agents follow same source control patterns as apps (and other solution components)
- You ~~can~~ should include agents in your ALM practices.
- You can see metadata and runtime in your Source Control.

Added Copilot Studio agent with Dataverse knowledge source

Browse F

10d98861 ODL_User 1926290 committed 2m ago

Files Details

Parent 1 → This commit | Filter

data
/PP/ContosoRealEstate/botcomponents/contoso_propertyFinderAssistant.gpt.default/data

odl_user_1926290

- PP/ContosoRealEstate
  - botcomponent_dvtablesearchs...
    - botcomponent_dvtable...
  - botcomponents
    - contoso_propertyFinderAssi...
      - botcomponent.yml
      - data
    - contoso_propertyFinderAssi...
      - botcomponent.yml
      - data
    - contoso_propertyFinderAssi...
      - botcomponent.yml
      - data
    - contoso_propertyFinderAssi...

```yaml
kind: GptComponentMetadata
instructions: |-
  - Assist real estate agents in finding suitable properties based on their requirements.
  - Prompt users to describe the property they are seeking, including location, type, and other relevant details.
  - Ask follow-up questions to clarify specific features such as number of bedrooms, bathrooms, amenities, and preferred neighborhood.
  - Request the user's budget to narrow down property options.
  - Provide suggestions or listings that match the described criteria and budget.
  - Maintain a professional and helpful tone throughout all interactions.
  - Do not provide legal, financial, or health advice.
  - Only respond in English and do not switch languages.
  - Avoid repeating information in consecutive messages.
  - Do not answer unrelated questions or generate creative content.
  - Decline requests that are harmful, offensive, or violate regulations.
gptCapabilities: {}
conversationStarters:
  - title: Find a property
    text: I'm looking for a property in New York with 3 bedrooms and a garden.

  - title: Set a budget
    text: My budget is $500,000. What options do I have?

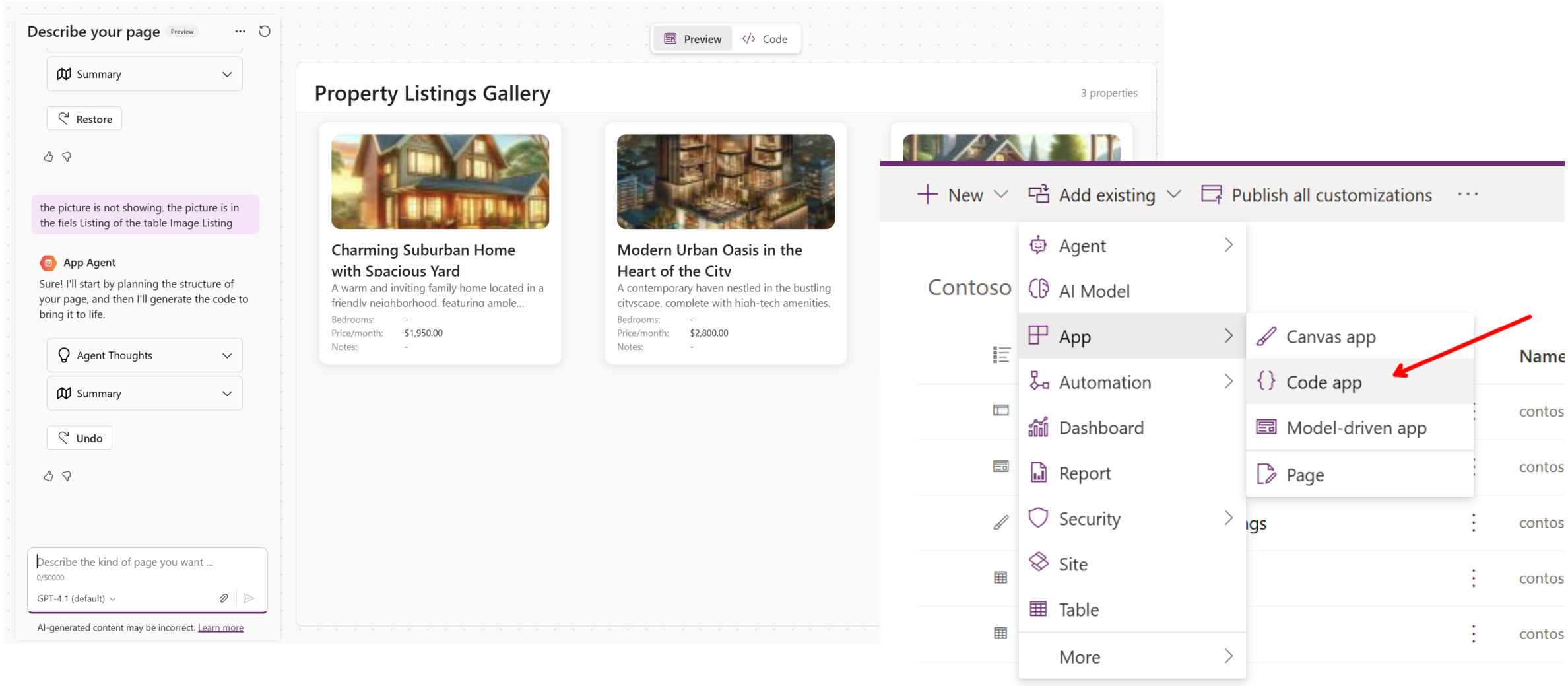
  - title: Describe features
    text: I need a property with a pool and close to schools.

  - title: Request amenities
    text: Show me properties with a garage and modern kitchen.

  - title: Location preference
    text: I prefer properties in downtown areas.

  - title: Compare listings
    text: Can you compare these two properties for me?
```

Generative Pages and Code Apps



Pro developer



You use PAC CLI / VSCode / Visual Studio



build PCF



build Plugins



Build web resources



Build Code apps

- For technical content: https://aka.ms/powerdev
- Agent academy: https://aka.ms/agent-academy

🔥 Topic for Pro devs 🔥

Code Apps: Easily Make Your React App Enterprise Grade with Power Apps

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Lunch 11:45 – 12:45

Agenda

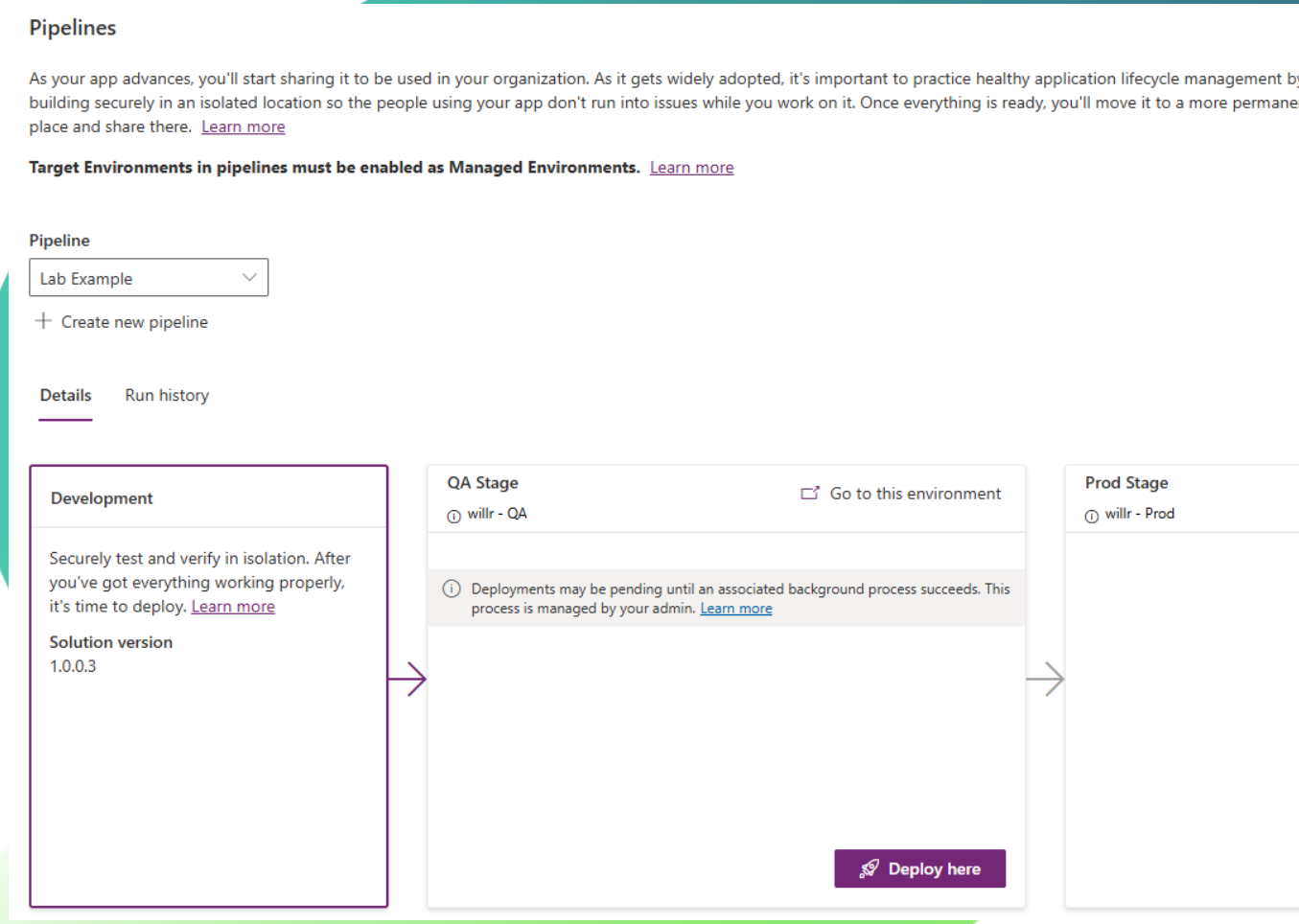
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Lab 3: Deploy

Power Platform Pipelines- In-Product ALM with Managed Environments

Pipelines in Power Platform offer an OOB advanced application lifecycle management (ALM) experience for Power Platform and Dynamics 365 by bringing ALM automation and continuous integration and continuous delivery (CI/CD) capabilities into the service in an approachable manner for all makers, admins, and developers.



Why use pipelines in Power Platform?

- Pipelines standardize solution movement from dev to test to production, reducing manual errors and ensuring predictable outcomes
- Common steps like export, import, and publish are automated, saving time and reducing operational overhead.
- They enforce environment security and approval workflows, aligning with organizational compliance requirements.
- Deployment logs and approval history provide transparency for troubleshooting and meeting audit requirements.
- Automatically stores managed and unmanaged versions of your solution for easy rollbacks– if needed.
- Included with Managed Environments

There are TWO types of pipelines

Platform Host

- Out of the box tenant-wide pipelines hosted under a “platform host” configuration
- These pipelines are configured by makers for **personal** use. Personal pipelines cannot be extended, nor can they be shared with other makers in the tenant

Custom Host

- Hosted on a “custom host” env installed and managed by admins
- Enable greater **control** as they are centrally governed by admins and shared with makers for use.
- Provide greater **visibility**, like audit where a certain solution is deployed and who initiated the request.
- Custom host pipelines also retain **backups** of solution artifacts in both Managed and Unmanaged forms.

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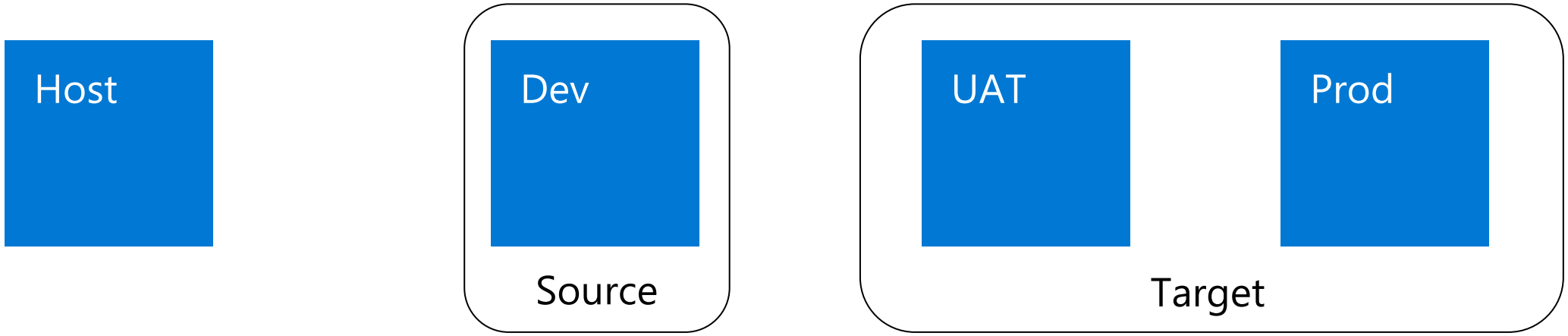
Create a Platform Host Personal Pipeline

Lab 3 – Task 1

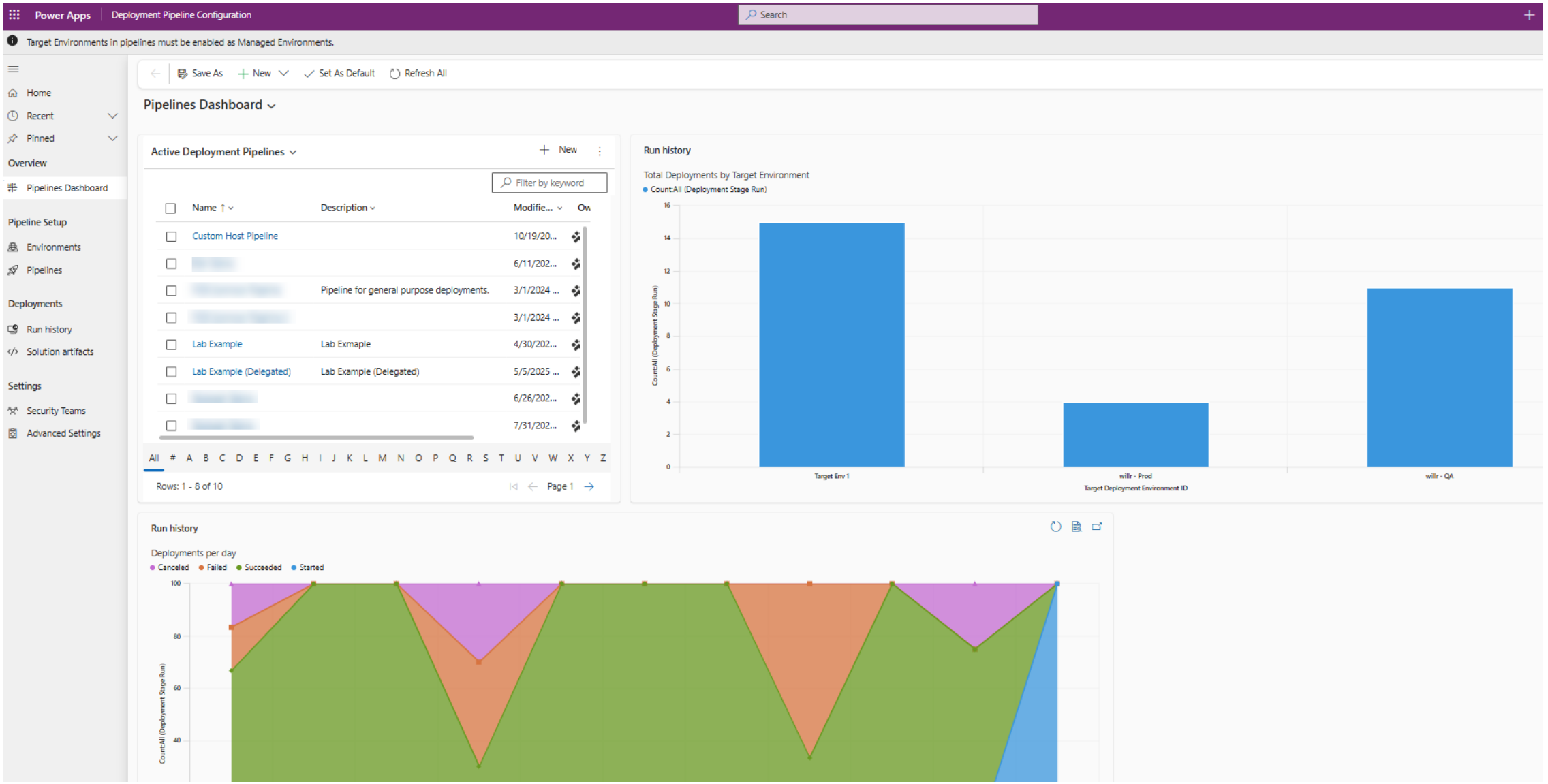
10 mins

Custom Host Pipelines

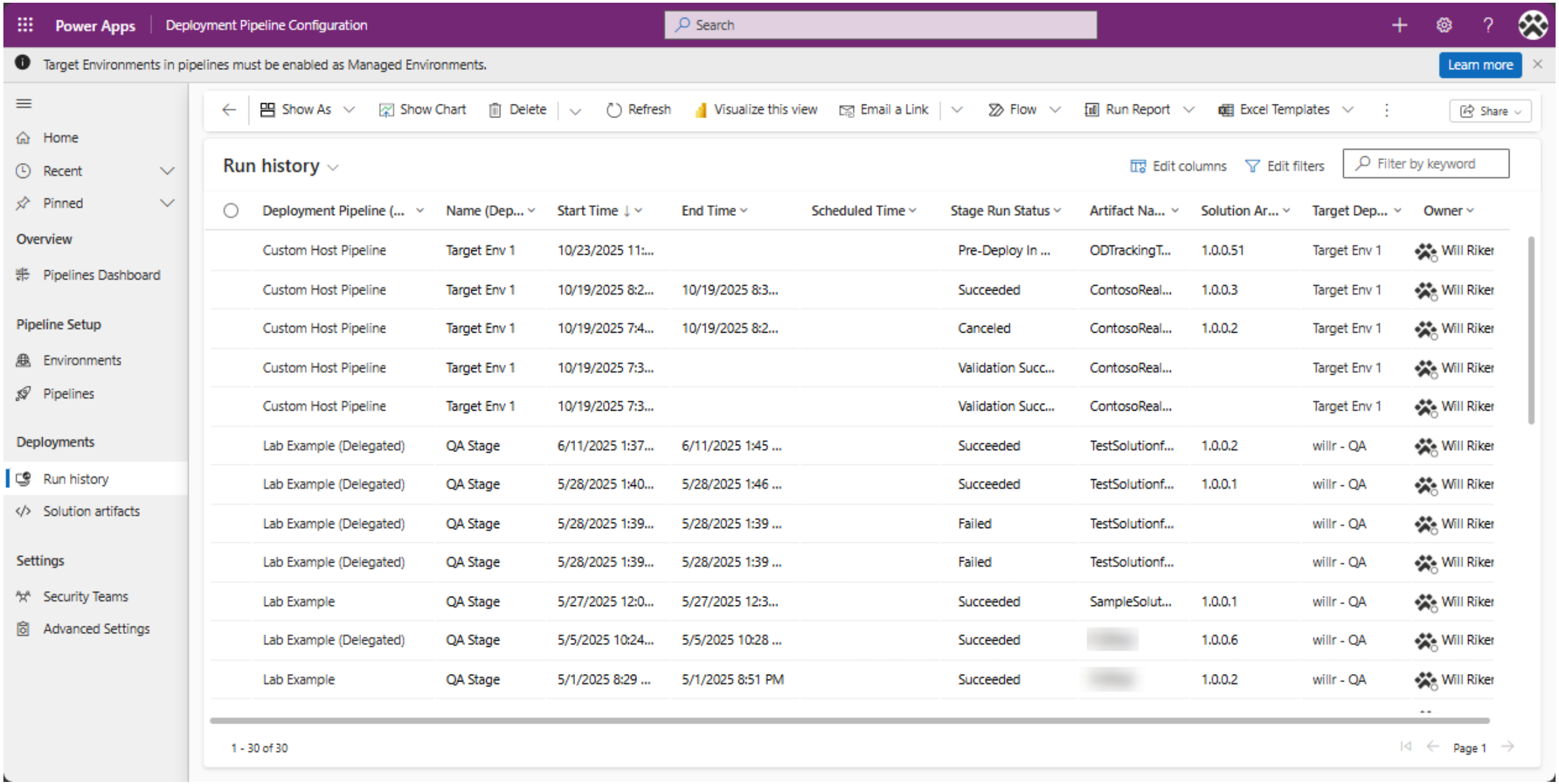
- Requires one dedicated Custom Host environment
- Managed from **Deployment Pipelines Configuration** app
- Supports up to **seven** deployment stages
  - At least one **Source** environment (usually Dev env)
  - Up to seven **Target** environments (UAT, Staging, Pre-Prod, Prod, etc.)



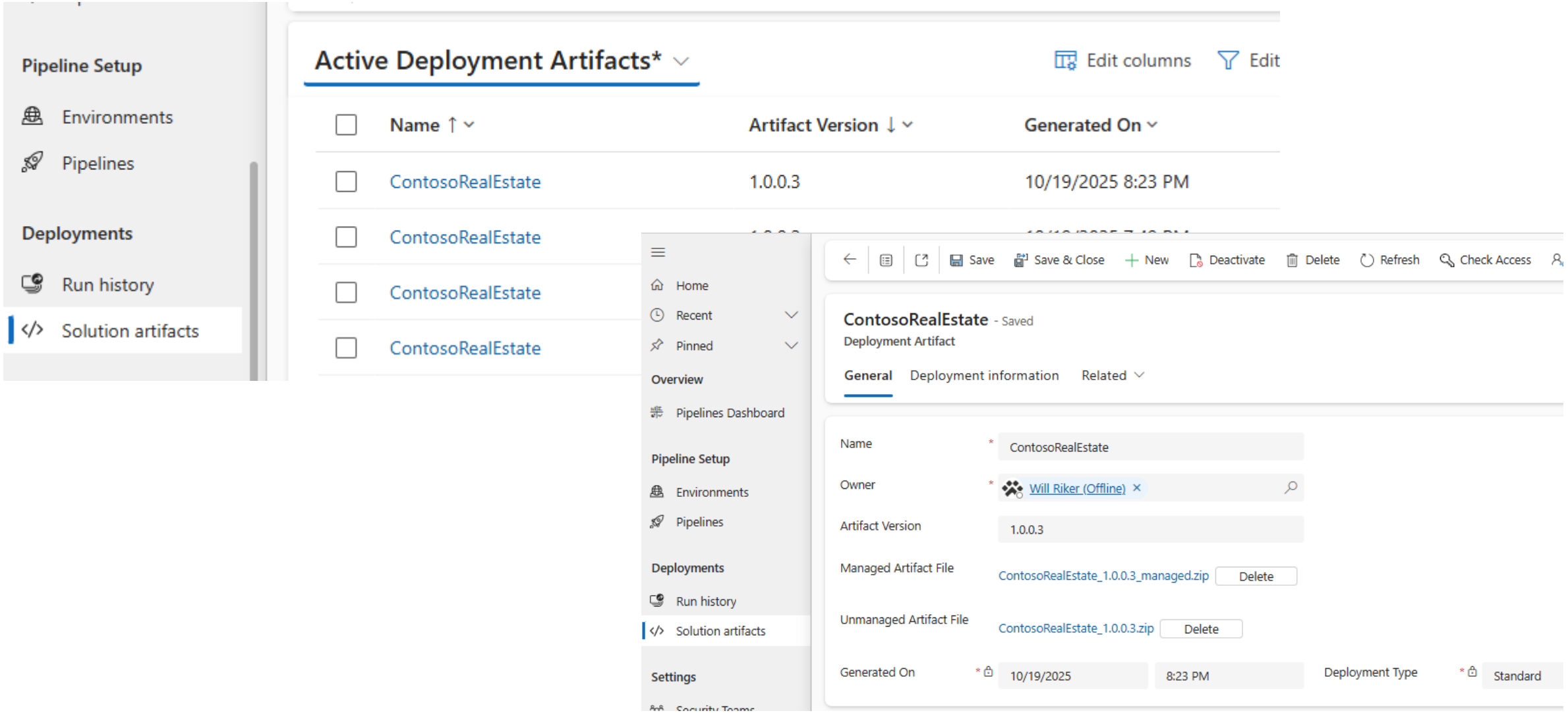
Deployment Pipeline Configurations app



Deployment Pipeline Configurations app



Deployment Pipeline Configurations app



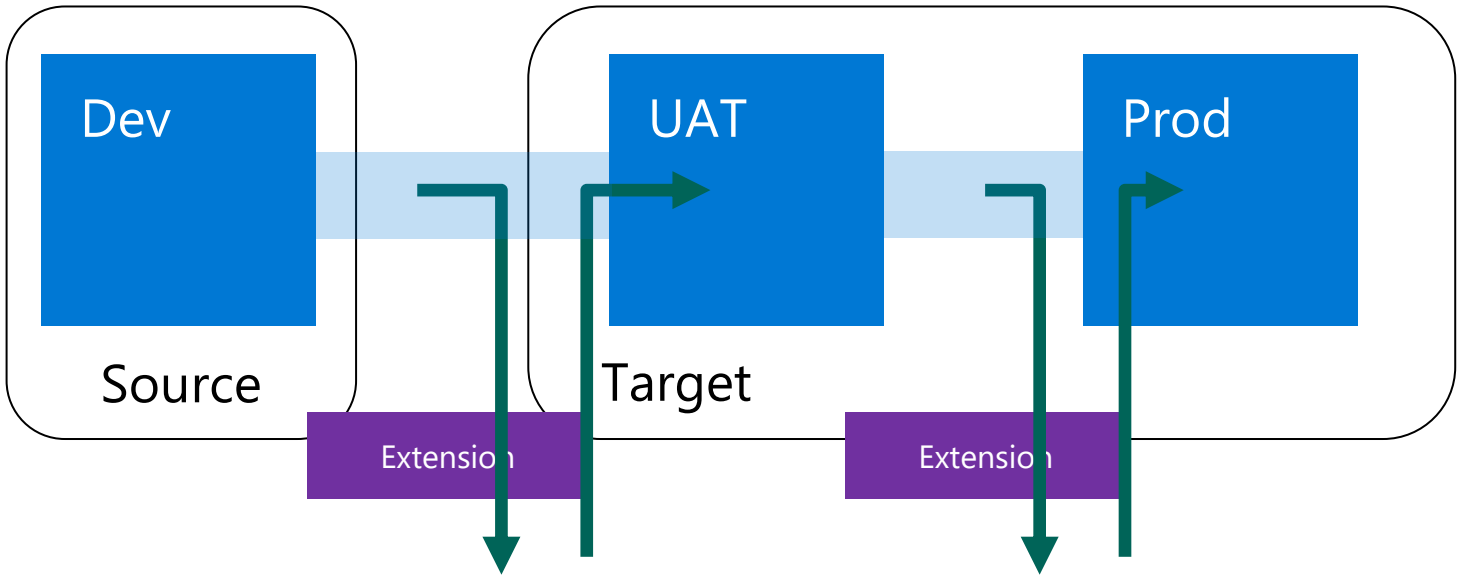
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Create a Custom Host Pipeline

Lab 3 – Task 2

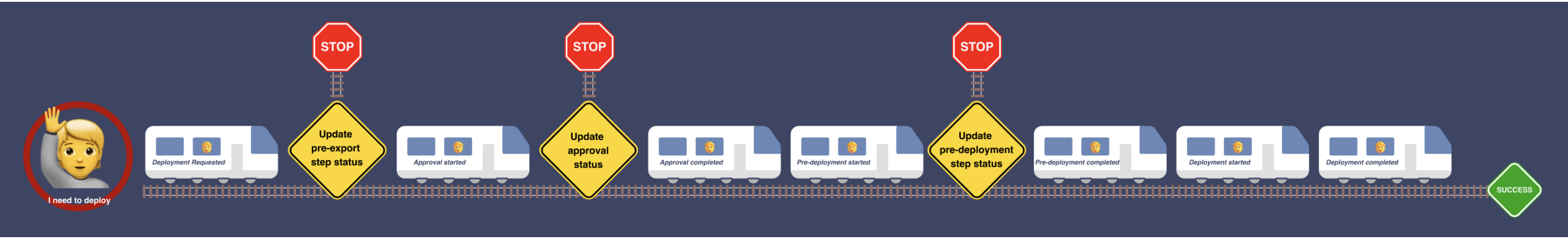
20 mins

Extending the Custom Host Pipeline

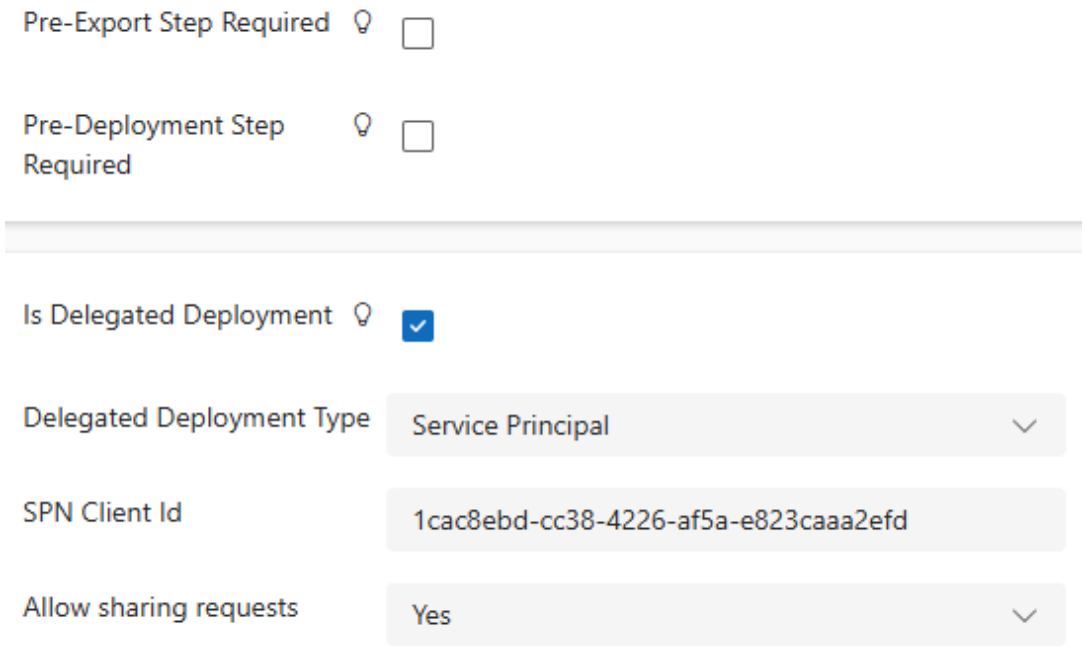




Power Platform Pipelines Extensibility

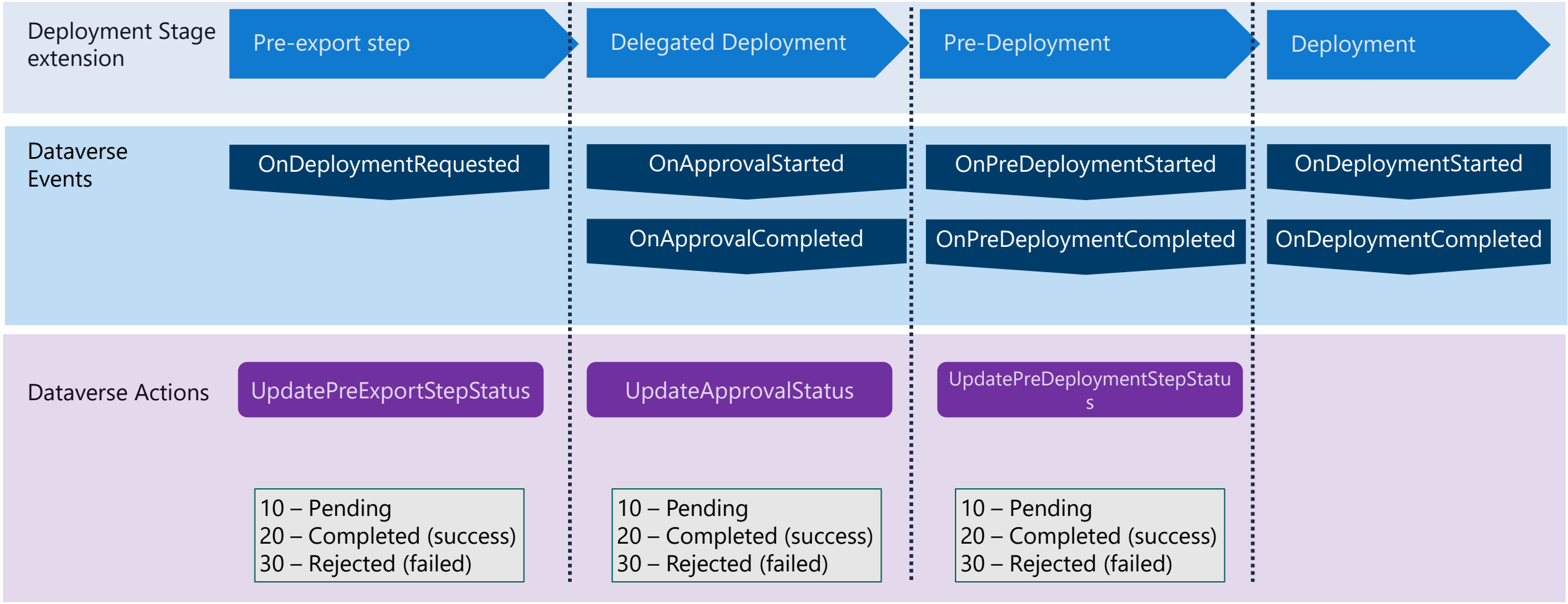


- Pipelines deployments progress through multiple predefined steps until deployment completion. Gated extensions insert custom steps into the progression where custom business logic can be executed.
- **Pre-export Step Required** allows running custom validation logic when a deployment request is submitted
- **Pre-deployment Step Required** provides additional flexibility to insert a custom step after a deployment is approved. For example, if a final approval is required
- **Is Delegated Deployment** carries out deployments using a service principal or pipeline stage owner's identity instead of the requesting maker's.



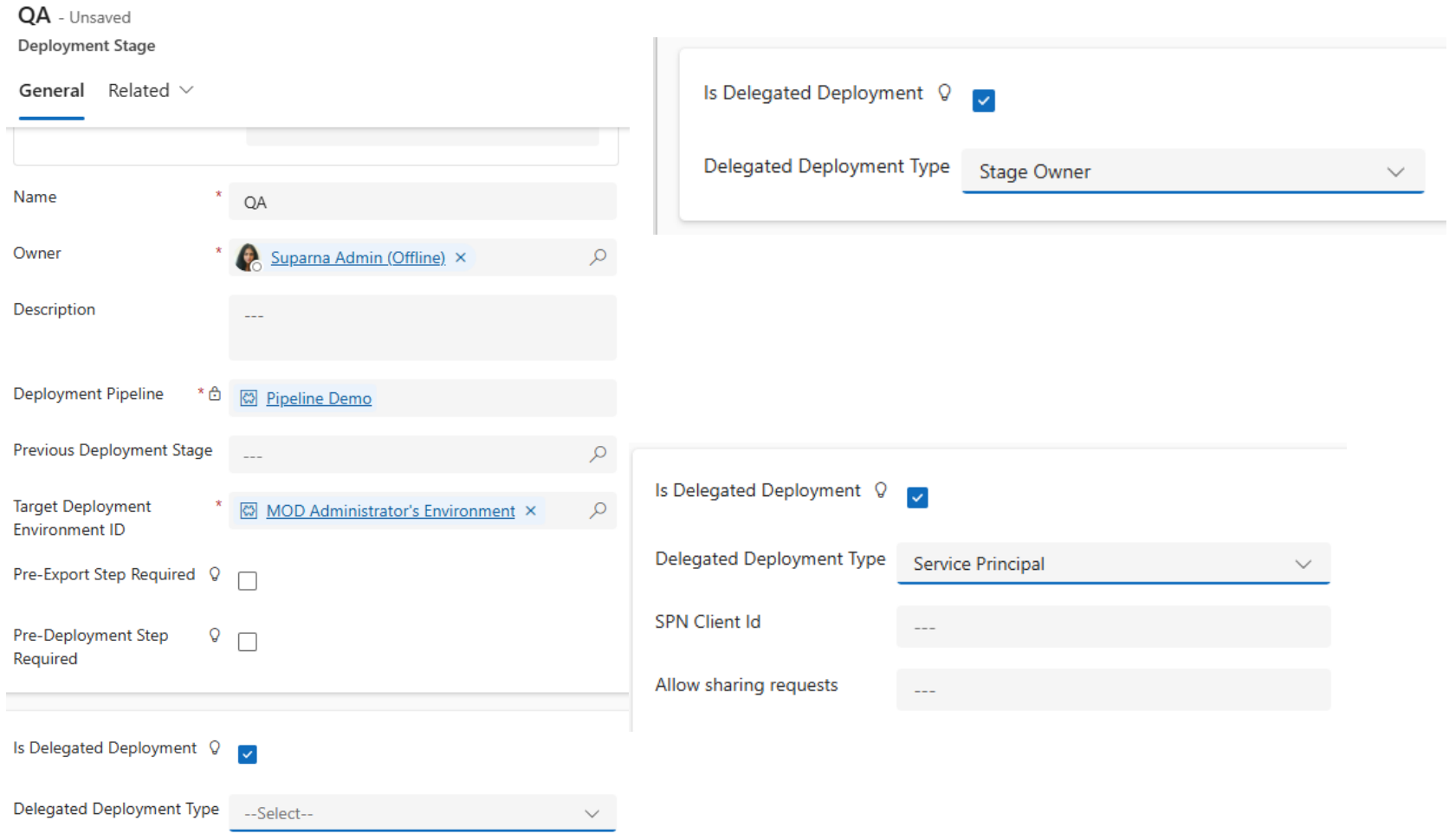
Extensions

- Each of these Extension events maps to a Dataverse event



Pipelines with Delegated Deployment

- Delegated deployments can be run as a service principal or pipeline stage owner.
- When enabled, the pipeline stage deploys as the delegate (service principal or pipeline stage owner) instead of the requesting maker.
- For deployment with Stage Owner, Configuration is more straightforward compared to service principals, but **solutions containing connection references for oAuth connections can't be deployed.**



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Extend Pipelines (Optional)

Lab 3 – Task 3

40 mins

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Redeploy a prior version

Lab 3 – Task 4

5 mins

Agenda

Lab	Time	Topic
Presentation	9:00 AM	Solutions and DevOps 101
Lab 1	9:20 AM	Setup and configure
Lab 2	9:40 AM	Building Power Platform Solutions
Coffee	**10:00 AM**	**Coffee break**
Lab 2 (cont)	10:15 AM	Building Power Platform Solutions
Lunch	**11:45 AM**	**Lunch**
Lab 3	12:45 PM	Deploy
Lab 4	2:15 PM	Lifecycle Mastery in Power Pages
Presentation	3:00 PM	Best practices & Q&A

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Lab 4: Lifecycle mastery in Power Pages

Agenda

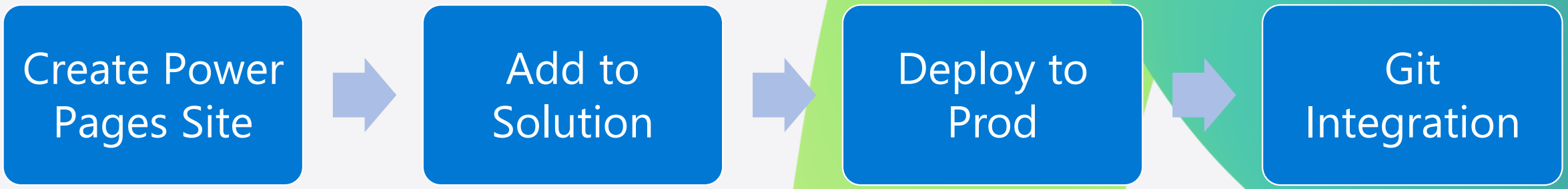
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Introduction

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Lab Overview



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DevOps Best Practices

Agenda

Lab	Time	Topic
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DevOps / ALM Best Practices for agents and other assets

Is DevOps different for agents vs other Power Platform assets?

No. While there are some Copilot Studio nuances, it's important all Power Platform assets can be managed with the same tools and processes.

DevOps will evolve over time, but the battle tested practices should remain consistent.

I’m new, what’s the first step I should take?

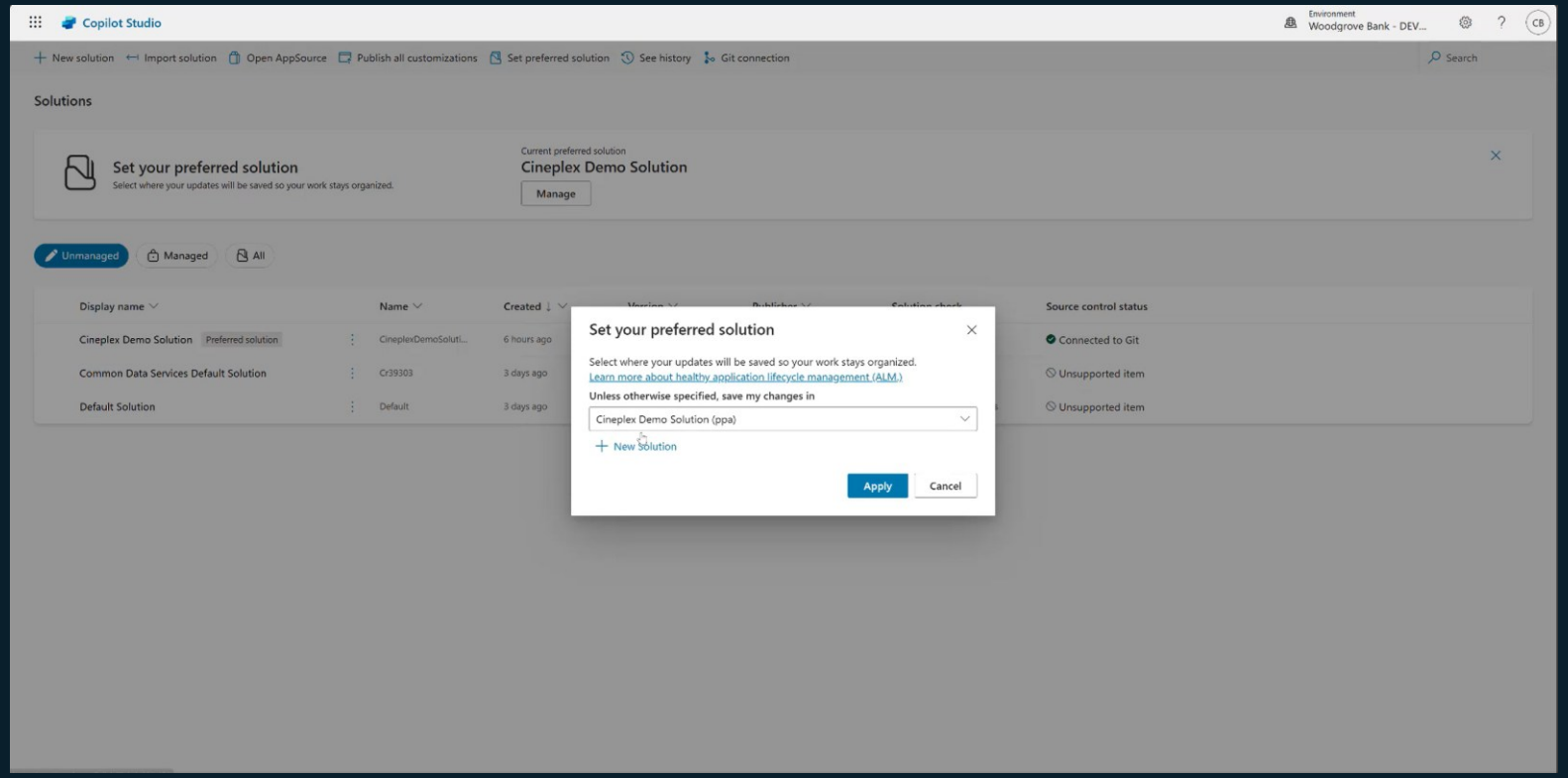
Use preferred solutions and Pipelines in Power Platform

Why? Anyone can be successful w/o understanding underlying complexities.

But… as you advance it's important to learn "how" the system works ☺

Set a Preferred Solution

- Avoids dependency issues downstream
- Ensures you don’t deploy someone else’s work inadvertently
- Essential for shared dev environments
- Per user, per environment



Use a single solution

*unless there is good reason not to!

Use a single solution publisher

Why? Publisher prefixes get stamped on everything created. They're immutable; if used improperly can cause layering issues.

A single publisher allows you to change your solution architecture later (which is common)

Deploy with a service principal

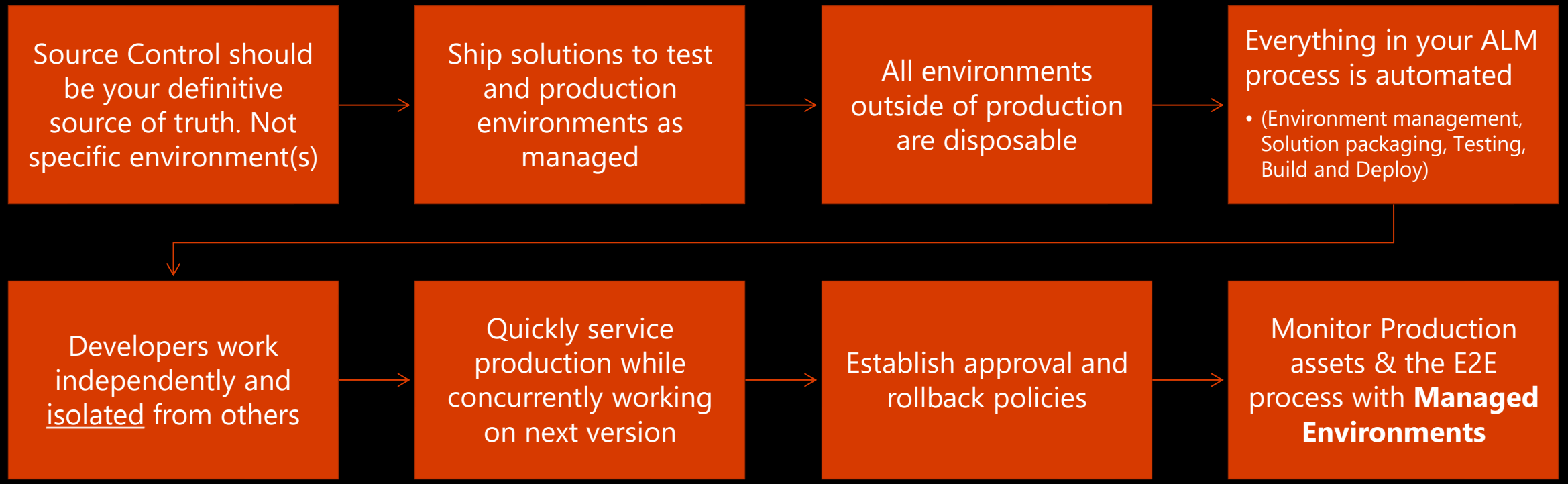
Otherwise, production environments are exposed to security risks

Use Solution Checker

Especially when deploying code

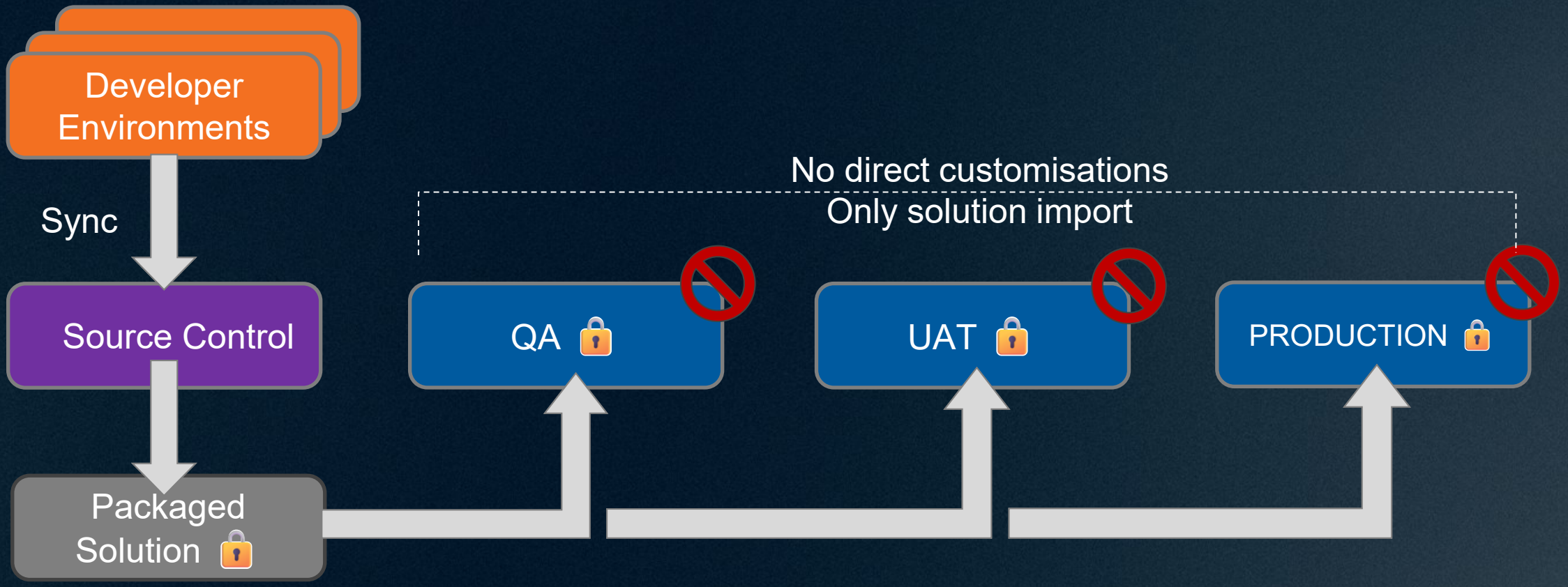
How do I manage team development effectively?

Healthy DevOps / ALM Overview



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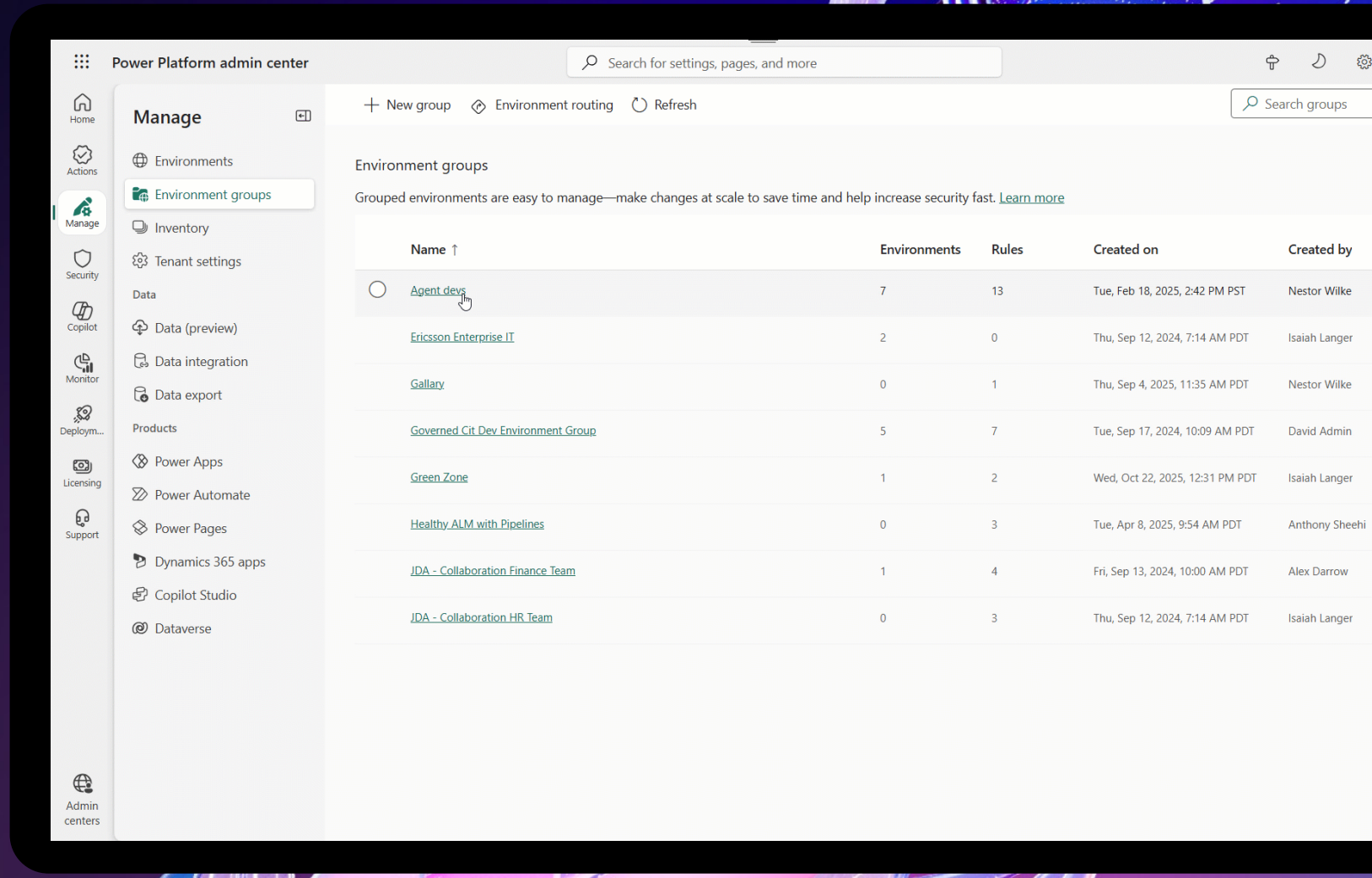
Source Control is the single source of truth
across time, versions, and environments



Default pipeline for groups + rules

Generally Available

- No more manual pipeline management
- Makers are automatically onboarded to the default pipeline
- Now Generally Available

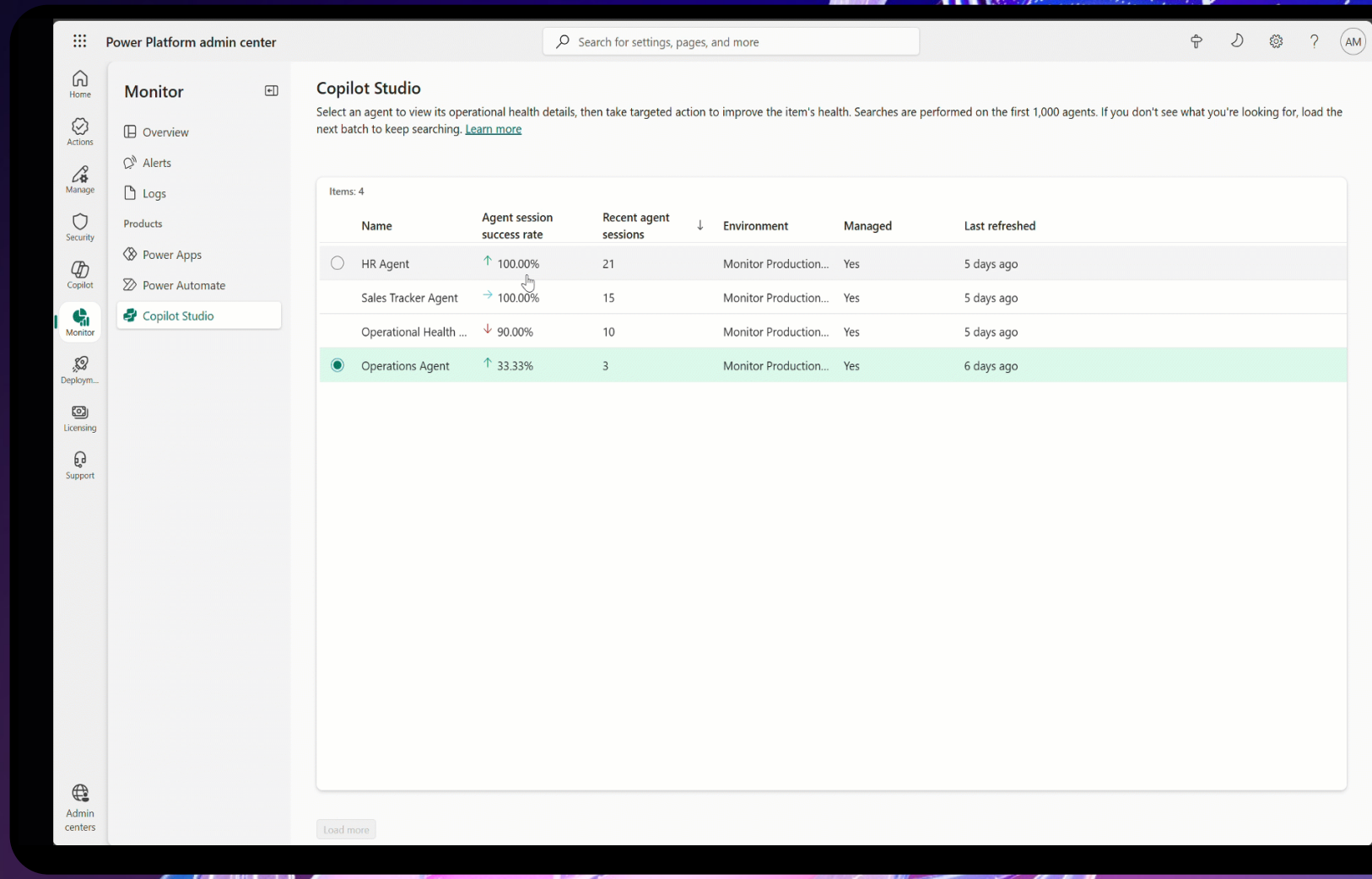


https://aka.ms/default-pipeline

Monitor health of Copilot Studio agents

Public preview

- **Monitor agent health** with the agent session success rate and recent agent sessions, now available in Monitor.
- Operations, CoE are now empowered to understand the health of agents.
- Minimize detecting agent degradations with alerts.



aka.ms/MonitoringHub

As an admin, how do I scale the process org-wide?

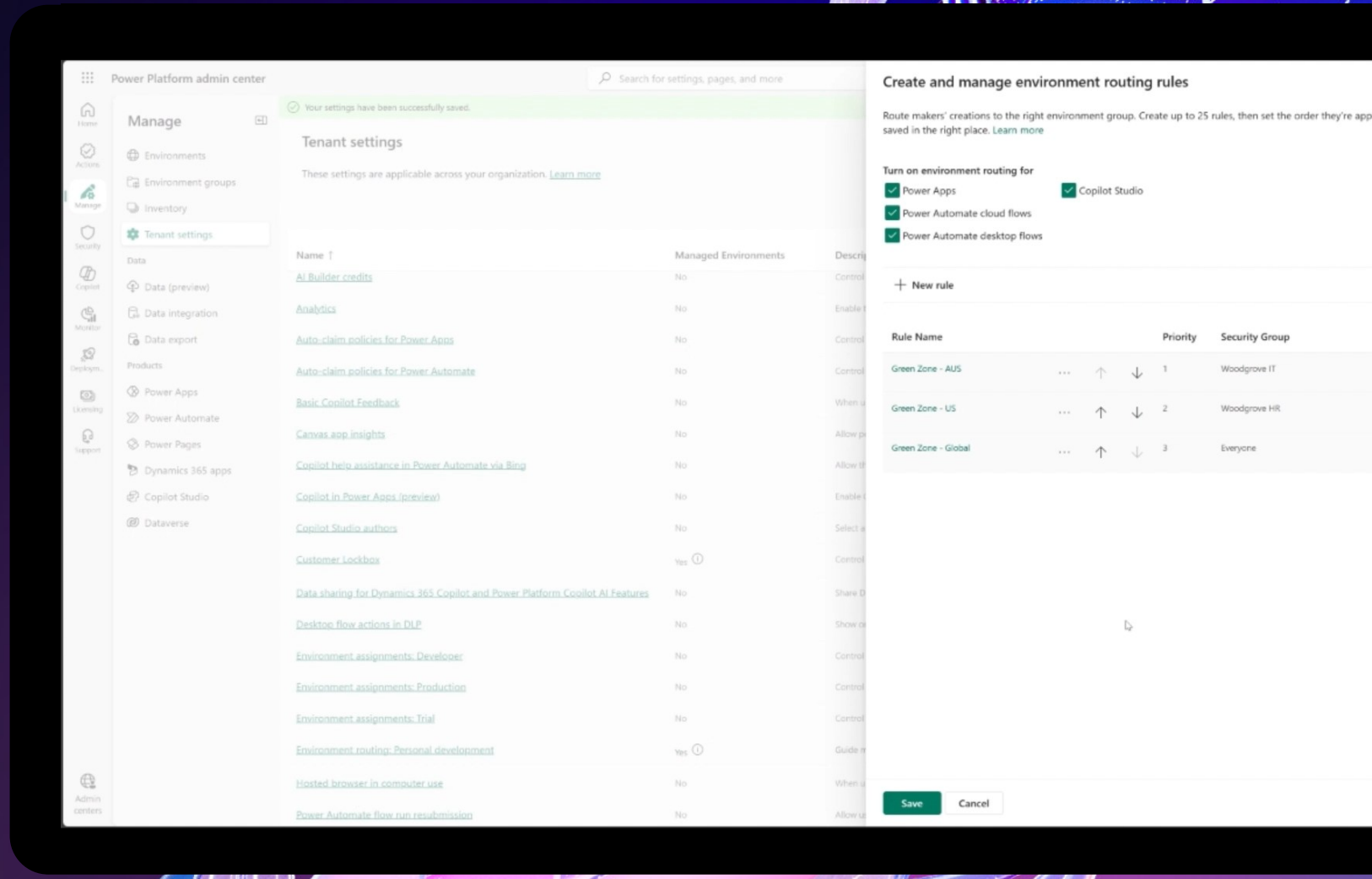
Environment routing

Generally available

Routing allows administrators to automatically direct users to personal developer environments **moving their makers away from the default environment**.

- Supports Copilot Studio, Power Apps, Power Automate Cloud and Desktop, and Flow Builder
- Designed to support simple to complex organizational structures.
- Attribute-based conditions via security groups or based on user properties such as geography, or department.
- Priority-based rule evaluation

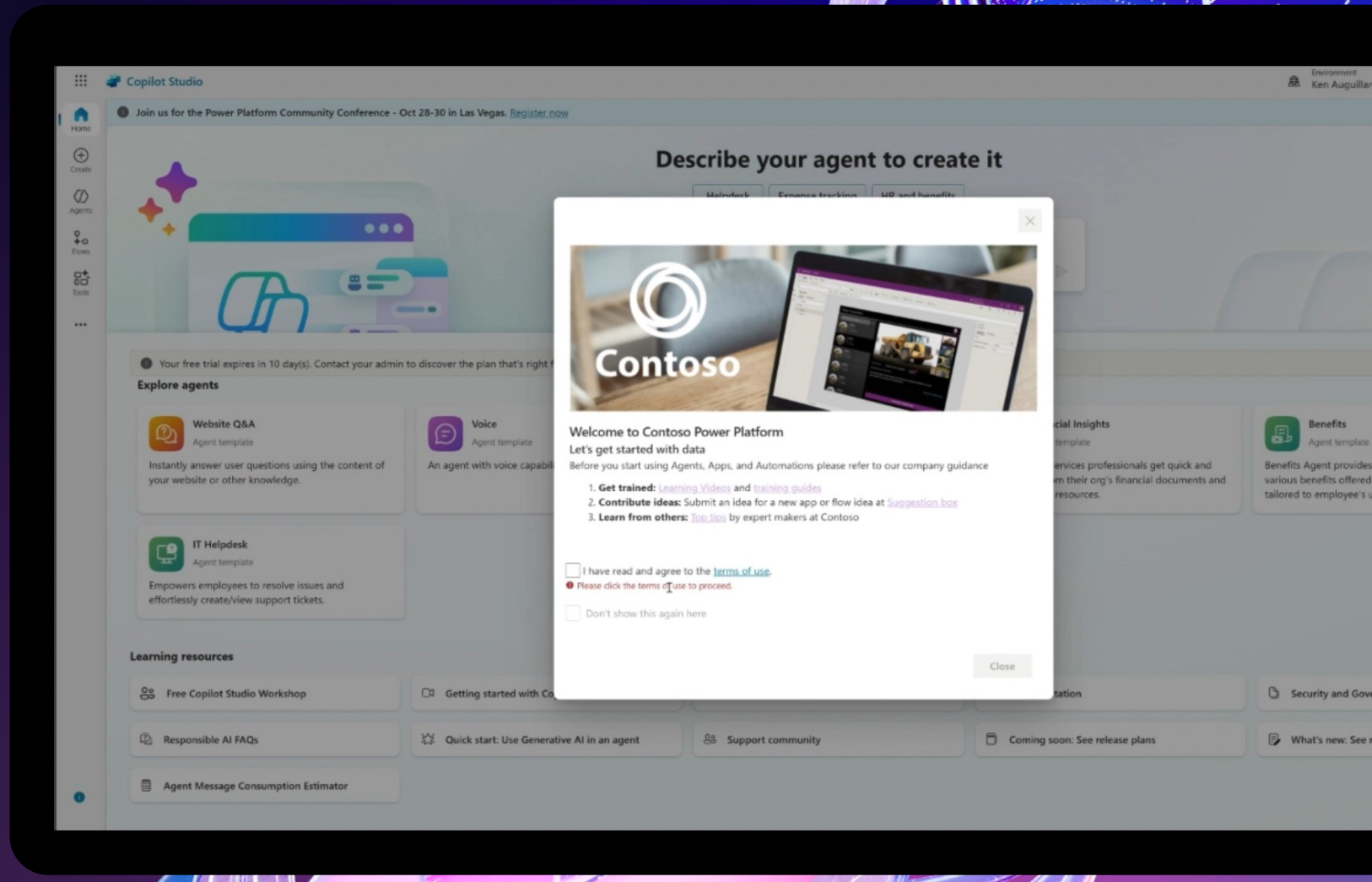
https://aka.ms/Environmentrouting



Maker onboarding

Public Preview (Nov. 31st)

- Centralized onboarding experience
- Ensures **consistent communication of policies, and guidelines** across environments.
- “Read and Understood” functionality — also known as: User Consent, or Terms of Acceptance.

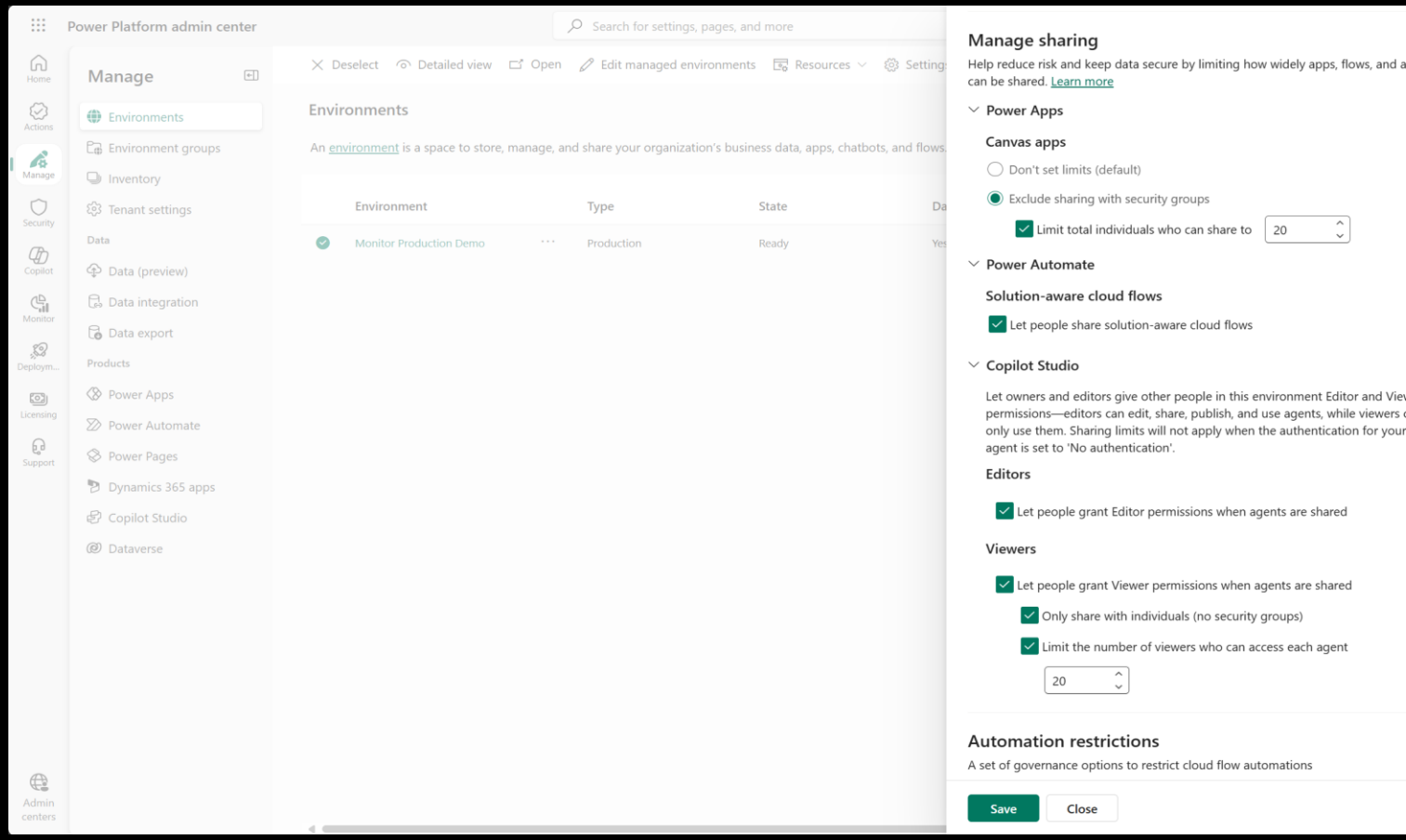


https://aka.ms/makeronboarding

Control how resources in your tenant are shared

Generally available

- Sharing limits give admins control over how resources are shared across the tenant—helping protect sensitive data from being exposed to the wrong audience.
- Now available for Copilot Studio agents and Power Apps code apps, in addition to canvas apps and cloud flows.

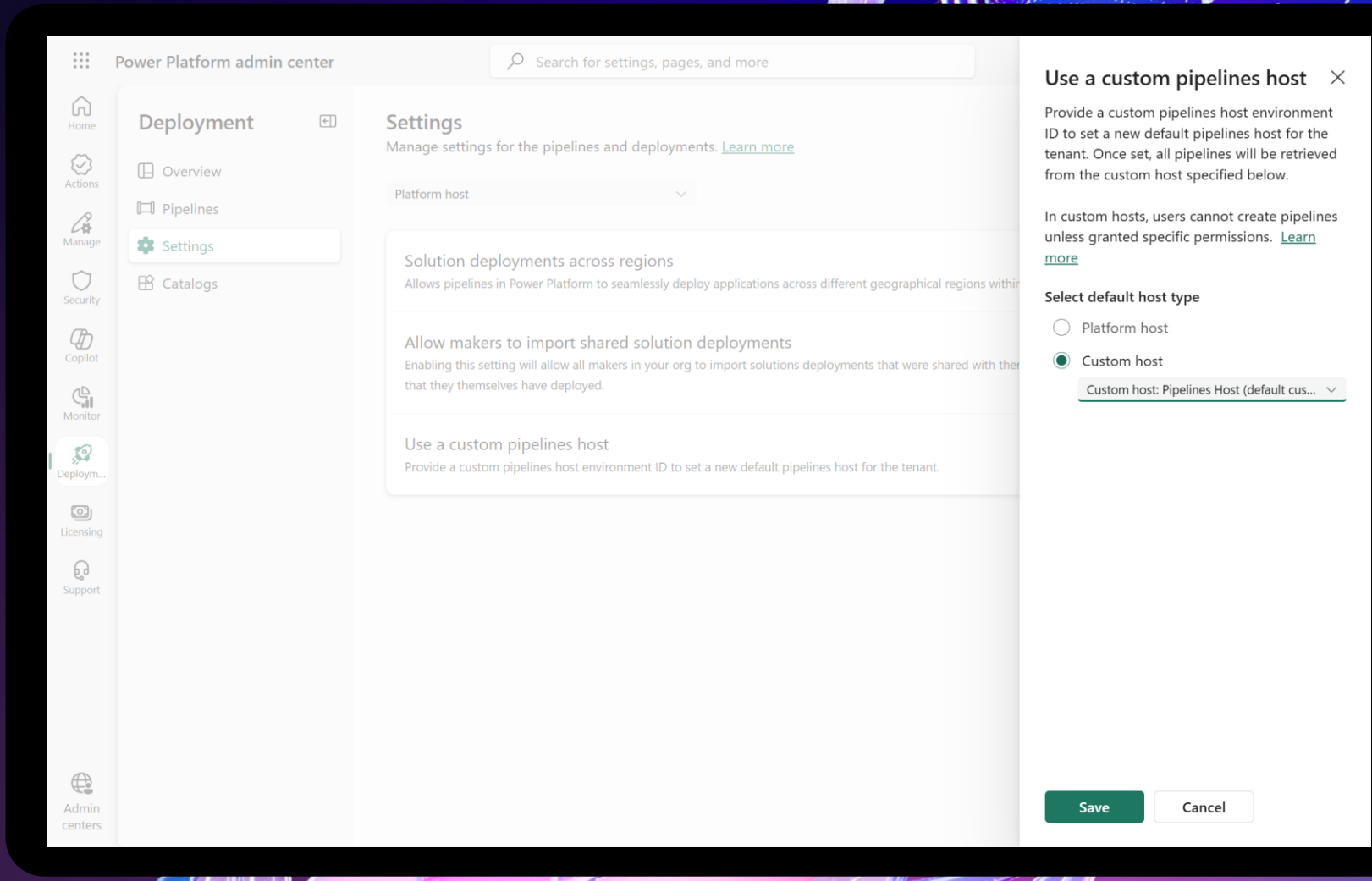


aka.ms/LimitSharing

Default custom host for Pipelines

Generally Available

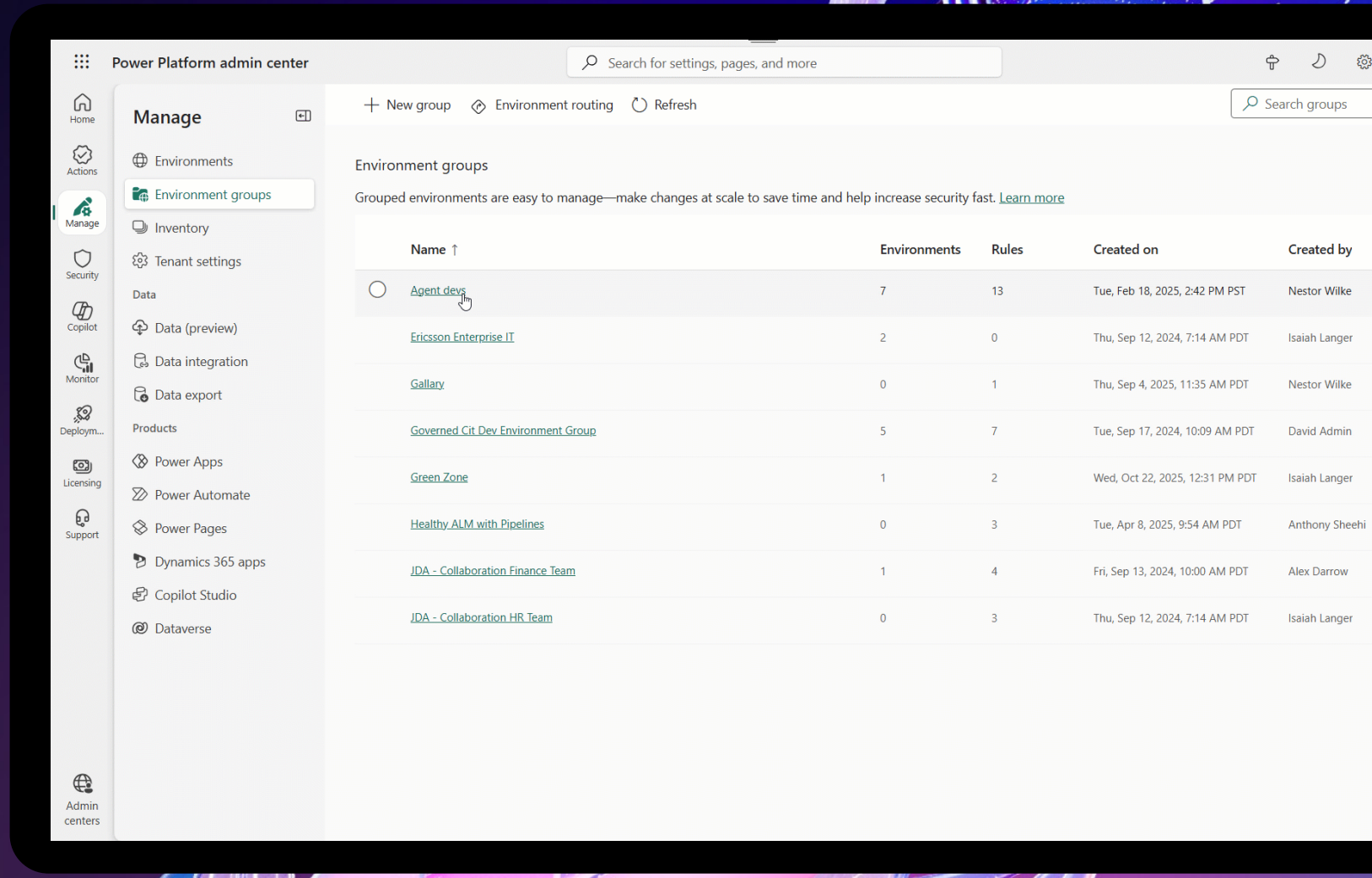
- Central Pipelines administration for the tenant
- Replaces the platform host with a custom host
- Full control over security and governance (unlike the platform host)



Default pipeline for groups + rules

Generally Available

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- Makers are automatically onboarded to the default pipeline
- Now Generally Available



https://aka.ms/default-pipeline

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Q & A

Microsoft Power Platform

Thank you

Join us Tues, Oct 28th at 2:15PM

MS250 Lifecycle Mastery: Managing Power Platform Assets in the Age of Agents

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